Expressions

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1. Expressions

1.1 Evaluating Expressions

A. Basics

1.1: Variables

- Numbers that we do not know can be represented using letters. These letters are also called variables, since their values can vary.
- Variables look complicated, but they are a simple of representing and working with information.

$$2+3=5$$
$$2\ Apples\ +3\ Apples=5\ Apples$$
$$2\ Dogs\ +3\ Dogs=5\ Dogs$$
$$2\ Dogs+3\ Apples=\ 2\ Dogs+3\ Apples$$

If there is some quantity that I do not know, I can represent that quantity with a letter.

$$2a+3a=5a$$

Some examples of variables are:

$$x,y,p,q,r$$

1.2: Assign Values to Variables

We can assign a value to a variable. This is often useful.

For example:

$$Cost\ of\ apple=a=2\$$$
$$Cost\ of\ orange=o=3\$$$
$$Cost\ of\ banana=b=\frac{1}{2}\$$$

B. Evaluating Expressions

Example 1.3

Find the value of the following expressions:

A. $a+3,\ a=7$
B. $12-y,\ y=2$
C. $3x,\ x=4$
D. $x+y$, if $x=3$ and $y=4$.
E. $2x+3y$ when $x=2$, and $y=4$
F. $2x+3y$, if $x=2$ and $y=10$.
G. $a+2b+3c$ when $c=3,b=2,a=1$

$$a=7\Rightarrow a+3=7+3=10$$

$$12-y,y=2\Rightarrow 10$$

$$3x,\qquad x=4\Rightarrow 3x=3\times x=3\times 4=12$$
$$x+y=3+4=7$$
$$2x+3y=2\times 2+3\times 4=4+12=16$$

$$2x + 3y = 34$$
$$a + 2b + 3c = 1 + 2 \times 2 + 3 \times 3 = 1 + 4 + 9 = 14$$

Example 1.4

Find which of the following expressions is greater when $a = 2, b = 3, c = 5$, and by how much:

$$A = 2a + 3b - c$$
$$B = 3a - b + 2c$$

$$A = 2a + 3b - c = 2 \times 2 + 3 \times 3 - 5 = 4 + 9 - 5 = 8$$

$$B = 3a - b + 2c = 3 \times 2 - 3 + 2 \times 5 = 6 - 3 + 10 = 13$$

$$13 > 8$$

$$Difference = 13 - 8 = 5$$

Hence, B is greater than A.

Example 1.5

Find the value of the following expressions if $x = 3, y = 5, z = 10$

A. $3x + 2y - z + 10$
B. $15 - x + y - z$
C. $22 + x + y - z$

$$3x + 2y - z + 10 = 3(3) + 2(5) - 10 + 10 = 9 + 10 = 19$$
$$15 - x + y - z = 15 - 3 + 5 - 10 =$$

Example 1.6: Fractions

Find the value of the following expressions:

A. $\frac{z}{5},\ z = 20$
B. $4x + 5y$, if $x = \frac{1}{2}$ and $y = \frac{2}{5}$.
C. $\frac{10}{x} + \frac{5}{y}$, if $x = 2$ and $y = 5$.

$$\frac{z}{5}, z = 20 \Rightarrow \frac{z}{5} = \frac{20}{5} = 4$$
$$4x + 5y = 4\left(\frac{1}{2}\right) + 5\left(\frac{2}{5}\right) = 2 + 2 = 4$$
$$\frac{10}{x} + \frac{5}{y} = \frac{10}{2} + \frac{5}{5} = 5 + 1 = 6$$

1.7: Order of Operations (BODMAS / PEMDAS)

We will perform mathematical operations in the following priority

- Brackets / Parentheses
- Exponentiation (Squares, cubes, or any power)
- Multiplication/Division
- Addition /Subtraction

$$3x^2 = 3 \times x^2$$
$$x^2(Exponentiation)\ has\ priority\ over\ the\ 3(multiplication)$$

Example 1.8: Squares

Find the values of the following:

Basics

A. $x^2 + y^2$, if $x = 10$ and $y = 5$
B. $2x^2 + 3y^2 + 4z^2$, if $x = 2, y = 1, z = 0$
C. $5a^2 + 2b^2 - z^2$, if $a = 1, b = 2, z = 3$

Fractional Values

D. $3p^2 - 2q^2$ if $p = \frac{1}{2}, q = \frac{1}{3}$

Evaluating Fractions

E. $\frac{12}{x^2} + \frac{1}{z^2}$ if $x = 4, z = 2$

Basics

$$x^2 + y^2 = 10^2 + 5^2 = 100 + 25 = 125$$

Taking squares has higher priority than multiplication:

$$2x^2 + 3y^2 + 4z^2 = 2(2^2) + 3(1^2) + 4(0^2) = 2(4) + 3(1) + 4(0) = 8 + 3 + 0 = 11$$

$$5a^2 + 2b^2 - z^2 = 5(1^2) + 2(2^2) - 3^2 = 5 + 8 - 9 = 4$$

Fractional Values

$$3p^2 - 2q^2 = 3\left(\frac{1}{2}\right)^2 - 2\left(\frac{1}{3}\right)^2 = 3\left(\frac{1}{4}\right) - 2\left(\frac{1}{9}\right) = \frac{3}{4} - \frac{2}{9} = \frac{27}{36} - \frac{8}{36} = \frac{19}{36}$$

Evaluating Fractions

$$\frac{12}{x^2} + \frac{1}{z^2} = \frac{12}{16} + \frac{1}{4} = \frac{3}{4} + \frac{1}{4} = \frac{4}{4} = 1$$

Example 1.9: Pythagorean Triplets

The value of $x^2 + y^2$ is equal to the square of some number. Find the number.

A. $x = 3$, and $y = 4$
B. $x = 6$, and $y = 8$
C. $x = 5$, and $y = 12$
D. $x = 8$, and $y = 15$

$$x^2 + y^2 = 3^2 + 4^2 = 3 \times 3 + 4 \times 4 = 9 + 16 = 25 = 5^2$$

$$B: x^2 + y^2 = 6^2 + 8^2 = 36 + 64 = 100 = 10^2$$

$$C: x^2 + y^2 = 5^2 + 12^2 = 25 + 144 = 169 = 13^2$$

1.10: Square Roots

Square root is the opposite of square. The symbol for square roots is the radical sign, written:

$$\sqrt{}$$

Example 1.11

A. $x = 36$ and $y = 100$. a is the square root of y, and b is the square root of x. Find $a - b$.
B. Find $\sqrt{x} + \sqrt{y}$ if $x = 1$ and $y = 81$
C. Find $\sqrt{x} + \sqrt{y}$ if $x = 100$ and $y = 0$

$$a = \sqrt{y} = \sqrt{100} = 10$$

$$a = \sqrt{x} = \sqrt{36} = 6$$
$$a - b = 10 - 6 = 4$$
$$\sqrt{x} + \sqrt{y} = \sqrt{1} + \sqrt{81} = 1 + 9 = 10$$

C. Real Life Scenarios

Example 1.12

An apple costs two dollars, and an orange costs three dollars. The expression $2x + 3y$ gives the cost of buying x apples and y oranges.

A. Mahesh goes to the market and buys 4 apples and 6 oranges. Find the money that he has to pay to the shopkeeper.
B. A grocery shop sells 100 apples and 200 oranges in the day. How much sales did he have?

$$2x + 3y = 2 \times 4 + 3 \times 6 = 8 + 18 = 26$$
$$2x + 3y = 2 \times 100 + 3 \times 200 = 200 + 600 = 800$$

Example 1.13

Troy works at a toy shop. He gets paid $7 per hour, and $1 for every toy that he sells. The expression $7h + t$ gives the money that he earns, where h is the number of hours worked, and t is the number of toys sold.

A. On Monday, Troy works three hours and sells five toys. How much money does he make?
B. On Tuesday, Troy works six hours and sells two toys for every hour that he has worked. How much money does he make?

$$7h + t = 7 \times 3 + 5 = 21 + 5 = 26$$
$$7h + t = 7 \times 6 + 12 = 42 + 12 = 54$$

Example 1.14

The expression $20c + 10$ gives the cost of a visit to the dentist, in dollars, where c is the number of cavities filled by the dentist. Dawn visits the dentist and get 3 cavities filled. How much must she pay?

$$No.of\ cavities = c = 3$$
$$20c + 10 = 20 \times 3 + 10 = 60 + 10 = 70\ dollars$$

Example 1.15

The number of shirts sewn by a tailor shop is given by $3a + 4b + 5c$, where a is the number of hours worked by Amar, b is the number of hours worked by Akbar, and c is the number of hours worked by Anthony. If Amar works 2 hours, Akbar works 3 hours, and Anthony works 4 hours, what is the number of shirts sewn?

$$3a + 4b + 5c = 3(2) + 4(3) + 5(4) = 6 + 12 + 20 = 38$$

Example 1.16

The cost of a garden shovel is $g - n$ dollars, where $g = 10$, and n is the number of shovels that you purchase.

A. Find the cost of purchasing five shovels.
B. Is there a problem with this formula?

We know that $g = 10$ and $n = 5$. Substitute these values in the given formula for cost of one shovel:

$$Cost\ of\ one\ shovel = g - n = 10 - 5 = 5\ dollars$$

Hence, the cost of five shovels is:

$$5 \times 5 = 25$$

Example 1.17: Formulas

A. Given the formula, $Speed = \frac{Distance}{Time}$ if a car travels a distance of 30 km in a time of 3 $hours$, then find the speed of the car.

B. Given the formula, $Time = \frac{Distance}{Speed}$ if a snail travels a distance of 20 $feet$ at a speed of $3\frac{ft}{hour}$, then find the time taken by the snail.

C. Given the formula, $Distance = Speed \times Time$ if the lava flow from a volcano moves at a speed of $3\frac{inches}{minutes}$ for a time period of 10 $hours$, then find the distance travelled by it.

Part A

$$Speed = \frac{Distance}{Time} = \frac{30\ km}{3\ hours} = 10\frac{km}{hr} = 10\ km\ per\ hour$$

Part B

$$Time = \frac{Distance}{Speed} = \frac{20\ ft}{3\frac{ft}{hour}} = \frac{20}{3}ft \times \frac{hour}{ft} = \frac{20}{3}hours$$

Part C

$$Distance = Speed \times Time = 3\frac{inches}{minutes} \times 600\ minutes = 1800\ inches = 150\ feet$$

A. Daisy Chaining Mathematical Operations

Example 1.18

A. A woman finds a magical pot. The pot doubles the number of fruits she keeps in it overnight. The woman starts by putting 5 fruits in it on Monday night. She eats 4 fruits every morning. How many fruits does she have on Friday night?

B. A donor offers to double the money in a fund at the end of every day. On Monday morning, 1 dollar gets collected. On Tuesday morning, 2 dollars gets collected. On Wed morning 3 dollars gets collected, and on Thu morning 4 dollars gets collected. How much money is available on Thu night?

Part A

$$\underbrace{5}_{\substack{Monday\\Night}} \to \underbrace{10}_{\substack{Tuesday\\Morning}} \to \underbrace{6}_{\substack{Tuesday\\Night}} \to \underbrace{12}_{\substack{Wed\\Morning}} \to \underbrace{8}_{\substack{Wed\\Night}} \to \underbrace{16}_{\substack{Thu\\Morning}} \to \underbrace{12}_{\substack{Thu\\Night}} \to \underbrace{24}_{\substack{Fri\\Morning}} \to \underbrace{20}_{\substack{Fri\\Night}}$$

Part B

$$\underbrace{1}_{\substack{Monday\\Morning}} \to \underbrace{2}_{\substack{Monday\\Night}} \to \underbrace{4}_{\substack{Tuesday\\Morning}} \to \underbrace{8}_{\substack{Tuesday\\Night}} \to \underbrace{11}_{\substack{Wed\\Morning}} \to \underbrace{22}_{\substack{Wed\\Night}} \to \underbrace{26}_{\substack{Thu\\Morning}} \to \underbrace{52}_{\substack{Thu\\Night}}$$

Example 1.19

A. A magical pot doubles the number of gold coins you put it in overnight. Niharika puts some gold coins in the pot on $5th$ night. She wakes up, and spends 10 gold coins. She puts the rest back in the pot on $6th$ night. On $7th$ morning, she finds 4 gold coins in the pot. Find the number of gold coins she had on $5th\ Night$.

B. A genie offers to double the number of oranges that an orange seller has. He does so in the morning, after which the seller sells 5 oranges. He then doubles the oranges in the afternoon, and the seller again sells 5 oranges. If the seller finishes with 9 oranges, find the number of oranges he had before he met the genie.

Part A

$$\underbrace{4}_{\substack{7th\\Morning}} \rightarrow \underbrace{2}_{\substack{6th\\Night}} \rightarrow \underbrace{12}_{\substack{6th\\Morning}} \rightarrow \underbrace{6}_{\substack{5th\\Night}}$$

Part B

$$\textit{Finished with } 9 \textit{ Oranges}$$
$$Before\ he\ sold\ 5\ Oranges\ in\ the\ afternoon = 14\ Oranges$$
$$Before\ the\ genie\ doubled\ the\ oranges = 7\ Oranges$$
$$Before\ he\ sold\ the\ 5\ oranges\ in\ the\ morning = 12\ Oranges$$
$$Before\ the\ genie\ doubled\ the\ oranges\ in\ the\ morning = 6\ Oranges$$

$$2(2x-5)-5=9$$
$$2(2x-5)=14$$
$$2x-5=7$$
$$2x=12$$
$$x=6$$

Example 1.20

A. Helen was counting ships in groups of fives. She divided her final answer for the number of groups by 5, and got 5. What was the number of ships she counted?

B. Arthur was asked to add $\frac{1}{3}$ to a number. Instead, he multiplied the number by $\frac{1}{3}$. If Arthur's answer was $\frac{2}{5}$, find the answer he should have got.

C. Gwen went lives in an apartment complex. She wanted to visit her friend, Merlin. Merlin told her, "My floor number is two floors below the floor with twice your floor number". Gwen instead went to the floor number that was twice of the floor that was two floors below her, and reached the tenth floor. What floor does Gwen live? Merlin?

Part A

$$No.of\ Ships \rightarrow \underbrace{\frac{No.of\ Ships}{5}}_{Groups\ of\ 5} \rightarrow \underbrace{\frac{No.of\ Ships}{25}}_{Dividing\ by\ 5} = 5$$
$$No.of\ Ships = 5 \times 25 = 125$$

Part B

$$No \times \frac{1}{3} = \frac{2}{5} \Rightarrow No. = \frac{2}{5} \div \frac{1}{3} = \frac{2}{5} \times 3 = \frac{6}{5}$$
$$No + \frac{1}{3} = \frac{6}{5} + \frac{1}{3} = \frac{18}{15} + \frac{5}{15} = \frac{23}{15}$$

Part C

$$Twice\ of\ \text{two floors below Gwen} = 10$$
$$Two\ floors\ below\ Gwen = 5$$
$$Gwen = 7$$
$$Twice\ of\ Gwen's\ floor = 14$$
$$Two\ floors\ below\ twice\ of\ Gwen's\ floor = 12$$
$$Merlin's\ floor = 12$$

Example 1.21
Seema was asked to add 12 to a certain number and then divide the result by 4. Instead, she first added 4, and then divided by 12, and ended up with 5 as the answer. If she had followed the instructions correctly, what would her answer have been?

Suppose that Seema started with a number. The process she followed is:

$$\underbrace{No.}_{\substack{Original\\Number}} \to \underbrace{No.+4}_{Add\ 4} \to \underbrace{\frac{No.+4}{12}}_{\substack{Divide\\by\ 12}} = 5$$

To find the original number, we need to do the opposite process:

$$\underbrace{12 \times \frac{No.+4}{12} = 5 \times 12}_{Multiply\ by\ 12} \to \underbrace{No.+4-4 = 60-4}_{Subtract\ 4} \to No. = 56$$

To find what her answer would have been if she had followed the process correctly, we go through all the steps:

$$\frac{56+12}{4} = \frac{68}{4} = 17$$

1.2 Writing Expressions: Basics

A. Addition

Till now we looked at using expressions that had been given to us to arrive at answers. Now, we look at writing our own expressions based on the given mathematical situation.
This skill is useful later on when setting equations.

1.22: Sum
Sum means to add

The sum of 6 and 8 is

$$6+8=14$$

Example 1.23
Convert the following statements into mathematical expressions. Use the small letter corresponding to the letter used for each part as a variable. For example, answer part A using a as the variable.

A. Seven added to a number
B. Five more than a number
C. A number added to ten
D. A number added to itself
E. The sum of six and a number
F. Farmer John has f apples harvested. He harvests 7 more apples. What is the number of apples that he has now?
G. Tiara has g cars with her. She gets a gift of three cars for her birthday. What is the number of cars that she has now?

Let the number be a. We are adding seven to a, hence the final answer is:

$$7 + \underbrace{a}_{Number} = 7 + a$$

$$b+5$$

$$c+10$$

$$d+d$$

$$e+6$$

$$f + 7$$
$$g + 3$$

1.24: Commutative Property
Order does not matter in addition:
$$a + b = b + a$$

B. Subtraction
A key difference between addition and subtraction is that order does not matter in addition. That is:
$$a + 5 = 5 + a$$

However, order does matter in subtraction. That is:
$$a - 5 \neq 5 - a$$

Because order is important, we must be careful with respect to which number comes first.

1.25: Difference
Difference means to subtract.

The difference between 7 and 10 is
$$10 - 7 = 3$$

1.26: Commutative Property
Order does not matter in subtraction.
$$a - b \neq b - a$$

- Subtraction does not satisfy the commutative property.

Example 1.27
You must be careful with subtraction. For example, compare
$$Addition: 3 + 4\ with\ 3 - 4$$
$$Subtraction: 3 - 4\ with\ 4 - 3$$

Addition remains the same irrespective of order:
$$3 + 4 = 4 + 3 = 7$$

Subtraction does not remain the same when you change the order:
$$3 - 4 = -1, \qquad 4 - 3 = 1$$

Example 1.28
In each case, decide which expression is correct:

A. Seven is subtracted from a number: $7 - a\ OR\ a - 7$
B. Four minus a number: $4 - b\ OR\ b - 4$

$$a - 7$$
$$4 - b$$

Example 1.29: Creating Expressions

Answer each question using the small letter corresponding to the capital letter used for the question number. For example, use the variable a in the answer to Part A.

A. Five less than a number.
B. A number decreased by twelve.
C. Three is subtracted from a number.

$$a-5$$
$$b-12$$
$$c-3$$

Example 1.30: Scenario Problems

A. A dog has f fleas. It goes to the veterinarian, who cleans the fur, and reduces the fleas by 5. Find the number of fleas the dog has after the visit.
B. I have 12 favourite plays that I want to see. However, my brother does not like p of those plays. If I only see plays that both of us like, how many plays can I see?
C. Tom takes a number and subtracts three from it. Find the value of the number after Tom has completed the subtraction. Use c for the number.

$$f-5$$
$$12-p$$
$$c-3$$

Example 1.31: Mixing Addition and Subtraction

A. The number of questions, q, that I attempted in the exam is five less than the total number of questions in the exam. Find the number of questions that I attempted. Also, find the total number of questions.
B.
C. The local municipality takes away three of the a mongrels in my locality. How many mongrels are left now?
D. There are 3 mongrels in my locality. The local municipality takes away a of them. How many mongrels are left now?

$$Number\ of\ questions\ attempted=q$$
$$Total\ Number\ of\ Questions=q+5$$

$$a-3$$
$$3-a$$

1.32: Keywords

Some keywords which are useful to note are:

$$Minus\ Sign: Subtraction, less, decreased, fewer$$
$$Plus\ Sign: Addition, more, increased$$

However, it is also important to note that the presence of a minus sign keyword does not always mean subtraction. The rest of the problem/logic matters.

Example 1.33: Keyword Check

A. I ate three apples fewer on Monday compared to Tuesday. On Monday, I ate m apples. How many apples did I eat on Tuesday?

B. Harish's age is 3 years less than his brother's age. If Harish is x years old, find his brother's age.

Part A

$$Apples\ on\ Monday\ are\ less \Rightarrow Apples\ on\ Tuesday\ are\ more$$
$$Apples\ on\ Monday = m$$
$$Apples\ on\ Tuesday = m + 3$$

Part B

$$x + 3$$

C. Multiplication

1.34: Notation

When we multiply a number with a variable we write:

$$3 \times x = x \times 3 = 3 \cdot x = x \cdot 3 = 3x$$

The convention in Algebra is to write the number before the variable and omit the multiplication sign:

$$3x\ and\ not\ x3$$

Example 1.35

A. $12 \cdot x$
B. $5\ times\ x$
C. $\frac{1}{3} \times 6 \times y$

$$12x$$
$$5x$$
$$\frac{1}{3} \times 6 \times y = \frac{6}{3}y = 2y$$

1.36: Product

Product means to multiply.

For example, product of 3 and 4 is

$$3 \times 4 = 4 \times 3 = 12$$

Example 1.37

Answer each question using a small letter corresponding to the capital letter in the question number. Use the variable a in the answer to Part A.

A. Three times a number.
B. The product of four and a number.
C. Seven multiplied by a number.
D. Twice of a number.
E. A number is tripled.
F. A number h is added to itself.
G. Twice of a number is added to the number itself.

$$3a$$
$$4b$$
$$7c$$

$$2d$$
$$3e$$
$$h + h = 2h$$
$$2g + g = 3g$$

1.38: Commutative Property

Order does not matter in multiplication.

$$a \times b = b \times a$$

Example 1.39

A. The baker has c cakes. He cuts each cake into four pieces to make pastries. What is the number of pastries that he has?
B. The baker has 4 cakes. He cuts each cake into c pieces to make pastries. What is the number of pastries that he has?

Part A

Originally, the baker had

$$c\ cakes$$

Then, he cut each cake into 4. Hence, the number of pastries is four times the number of cakes:

$$4 \times c = 4c$$

Part B

Originally, the baker had

$$4\ cakes$$

Then, he cut each cake into c parts. Hence, the number of pastries is c times the number of cakes:

$$c \times 4 = 4c$$

Example 1.40

A. Rehan had x dollars in his piggy bank. Then his mother doubled the number of dollars he had. What is the money that he has now?
B. Every dollar has 100 cents. Sheetal has y dollars. What is the number of cents she has?

$$x + x = 2x\ dollars$$
$$100y\ cents$$

D. Division

1.41: Division Statement-I

$$\frac{Dividend}{Divisor} = Quotient + \frac{Remainder}{Divisor}$$

$$\frac{21}{4} = 5 + \frac{1}{4}$$

$$\frac{\underbrace{21}_{Dividend}}{\underbrace{4}_{Divisor}} = \underbrace{5}_{Quotient} + \frac{\underbrace{1}_{Remainder}}{\underbrace{4}_{Divisor}}$$

1.42: Division Statement-II

$$Dividend = Quotient \times Divisor + Remainder$$

$$21 = 5 \times 4 + 1$$
$$\underbrace{21}_{Dividend} = \underbrace{5}_{Quotient} \times \underbrace{4}_{Divisor} + \underbrace{1}_{Remainder}$$

1.43: Quotient
Quotient means to divide and find the "answer".

$$Quotient\ of\ a\ and\ b = a \div b = \frac{a}{b}$$

For example, the quotient of 13 and 5 is given by:

$$\frac{13}{5} = 2 + \frac{3}{5} \Rightarrow 2\ is\ Quotient, 3\ is\ the\ Remainder$$

Note that the first number (13) goes in the numerator, and the second number (5) goes in the denominator.

1.44: Quotient does not satisfy commutative property
The order in which we undertake the division is important.

$$\frac{a}{b} \neq \frac{b}{a}$$

For example, sharing two apples among ten people is not the same as sharing ten apples among two people.

1.45: Fractions as division
You can represent division as fractions

$$a \div b = \frac{a}{b}$$

Example 1.46
Answer each question using the small letter corresponding to the capital letter used for the question number. For example, use the variable a in the answer to Part A.

A. The quotient of seven and a number
B. Five divided by a number
C. The quotient of a number and three
D. A number divided by twelve

$$7 \div a = \frac{7}{a}$$
$$5 \div b = \frac{5}{b}$$

Check the difference between this question and the previous one. In the previous question, five is divided by a number. Hence, the variable goes in the denominator. In this question, a number is divided by three. Hence, the variable goes in the numerator.

$$c \div 3 = \frac{c}{3}$$
$$\frac{d}{12}$$

Example 1.47

Farmer John harvested p apples from his orchard. He then divided the apples into three equal piles. Find the number of apples in each pile.

Farmer John starts with p apples.
He then divides them into three equal piles, which gives him

$$p \div 3 = \frac{p}{3}\ Apples\ in\ each\ pile$$

1.3 Writing Expressions-II

A. Working with Totals

1.48: Working with Totals

Often, we need to work with quantities divided into two types:

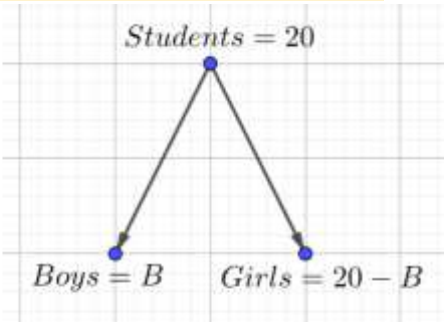
- Students in a class can be either *boys* or *girls*
- You can either *pass* or *fail* in a subject
- A fruit can be *good* or *rotten*

In this scenario, if you have the total number, you do not need to take two variables, since the second quantity will be the

$$Total - First\ Quantity$$

Example 1.49

There are 20 students in a class. There are B boys. Find the number of girls.



Since the number of boys and girls adds up to the students, the number of girls is determined by subtracting the number of boys from the total number of students:

$$Girls = G = 20 - B$$

Example 1.50

I have 12 fruits (either apples or oranges) with me. If I have A apples, what is the number of oranges?

$$Oranges = O = 12 - A$$

Example 1.51

I have 7 dollars which I spend on food and books. I spend f dollars on food. How much do I spend on books?

$$Books = b = 7 - f$$

Example 1.52

There are R Republicans on a Senate Committee with seventeen members. If the committee has only Republicans and Democrats, what is the number of Democrats?

$$Democrats = D = 17 - R$$

Example 1.53

A dragon lair has two types of dragons: gold and red. There are a total of 19 dragons in the lair. If there are G gold dragons, what is the number of red dragons?

$$Red\ Dragons = R = 19 - R$$

B. Multi-Step Questions

Example 1.54

There are x people at a beach on Friday. On Saturday, twelve more people come to the beach as compared to Friday. On Sunday, the number of people is double of the number of people on Friday. The number of people on the beach on Monday was double of the number of people on the beach on Saturday. Find the number of people, in terms of x, on the beach on:

A. Friday
B. Saturday
C. Sunday
D. Monday

$$Friday: x$$
$$Saturday: x + 12$$
$$Sunday: x \times 2 = 2x$$
$$Monday: 2(x + 12) = 2x + 24$$

Example 1.55

Jake (x years old) is five years older than his brother, John (who is y years old).

A. Find Jake's age in terms of John's age
B. Find John's age in terms of Jake's age.
C. What is the smallest value that x can take?

Part A

John's age is $y\ years$. And Jake is five years older than John. Hence, his age is:

$$y + 5$$

Part B

Jake's age is $x\ years$. And Jake is five years older than John. Hence, John is five years younger than Jake. Hence, his age is:

$$x - 5$$

Part C

Minimum value of John's age is

$$0\ Years$$

And hence, minimum value of Jake's age is

$$0 + 5 = 5\ Years$$

Example 1.56

A. There are p people at a beach on Friday. On Saturday, five more people come to the beach as compared to Friday. On Sunday, the number of people is double of the number of people on Saturday. Find the number of people, in terms of p, on the beach on Sunday.
B. Yesterday, I met p people for work. Today, I met five more people compared to yesterday. What is the number of people I met today?
C. Yesterday, I met p people for work. Yesterday, I met seven more people compared to today. What is the

number of people I met today?
D. I met c children on my way to work yesterday. Half of them were boys. How many boys did I meet?
E. The time now is m minutes after noon. I just completed lunch, which took me 30 minutes. What was the time when I started lunch?
F. I get into a bus at y minutes after 4 pm. The bus journey takes 17 minutes. What time do I get off the bus?

Part A

$$Friday: p$$
$$Saturday: p+5$$
$$Sunday: p+p+10 = 2p+10$$

Part B

$$\underbrace{p}_{Yesterday}, \underbrace{p+5}_{Today}$$

Part C

$$\underbrace{p}_{Yesterday}, \underbrace{p-7}_{Today}$$

Part D

$$\frac{c}{2}, \quad c \text{ is even}$$
$$c = 5$$

Part E

$$Current\ time = m\ minutes\ after\ noon$$

Time when he started lunch will be before the time that he finished lunch:

$$Started\ time = (m-30)\ minutes\ after\ noon$$

Part F

$$Get\ into\ the\ bus\ at\ 4pm + y\ minutes$$
$$Get\ off\ the\ bus: 4\ pm + (y+17)\ minutes$$

Example 1.57

A. A baker takes c minutes to make a cake. He then takes p minutes to cut the cake into three pastries. In how much time will he make three cakes and one pastry?
B. Dhruva takes three times longer to finish his homework than his brother. Dhruva takes h minutes to finish his homework. How long does his brother take?
C. I want to buy a crate of bananas. Each banana costs Rs. 3. I have to pay Rs. c for the crate separately. If the crate has b bananas, what is the cost of buying the bananas (with the crate)?
D. y burgers were sold yesterday, which is twice the number of burgers sold today. What is the number of burgers sold today?
E. Nalini is three times as old as her brother. Nalini is n years old. How old is her brother?
F. A library has a collection of b rare books. A benefactor donates 3 books to the library. And 7 books are taken away for an exhibition. What is the number of books in the library now?
G. Hetal has c cakes. She cuts each cake into two slices. Then, she gets three more slices of cake. What is the number of slices she has now?

Part A

$$1\ Cake = c\ minutes \Rightarrow 3\ Cakes = 3c\ minutes$$

$$3\ Pastries = p\ minutes \Rightarrow 1\ Pastry = \frac{p}{3}$$

$$Total = 3c + \frac{p}{3}$$

Part B

$$Dhruva\ takes\ h\ minutes$$

His brother takes less than time

$$Brother\ takes\ \frac{h}{3} minutes$$

Part C

$$Cost\ of\ one\ banana = 3$$
$$Cost\ of\ b\ bananas = 3b$$
$$Cost\ of\ crate = c$$

Total Cost:

$$= \underbrace{3b}_{\substack{Cost\ of \\ Bananas}} + \underbrace{c}_{Crate}$$

Part D

$$Burgers\ sold\ yesterday = y$$

Burgers sold today are half the number of burgers sold yesterday. Hence:

$$Burgers\ sold\ today = \frac{y}{2}$$

Part E

$$Nalini's\ age = n\ years$$

Her brother is one-third Nalini's age. Hence, his age is:

$$\frac{n}{3} years$$

Part F

$$b + 3 - 7 = b - 4$$

Part G

$$2c + 3$$

Example 1.58

The number of people who visited the theatre yesterday is twelve more than the number of people who visited the theatre today. The number of people going to visit the theatre tomorrow is twelve less than the number of people who visited the theatre today. p people visited the theatre yesterday. What is the total number of people visiting the theatre across yesterday, today and tomorrow?

$$\underbrace{p}_{Yesterday} + \underbrace{p-12}_{Today} + \underbrace{p-24}_{Tomorrow} = 3p - 36$$

Example 1.59

A. A farmer harvests half of t apples from a tree and then a bird eats two apples from the harvested ones? In the same farm, a bird eats two mangoes from a tree with t mangoes. Then, the farmer harvests half the remaining mangoes. How many apples and mangoes are with the farmer?

B. Sonam is three years older than her brother. She is also twice as old as her brother. Write Sonam's age in two different ways, if her brother's age is b.

C. Tvisha is six years younger than her brother. She is also half as old as her brother. Tvisha is t years old. Find her brother's age in terms of Tvisha's age, in two different expressions.

Part A

$$Apples: t \underbrace{\rightarrow}_{\substack{Divide \\ by\ 2}} \frac{t}{2} \underbrace{\rightarrow}_{\substack{Subtract \\ 2}} \frac{t}{2} - 2$$

$$Mangoes: t \underbrace{\rightarrow}_{\substack{Subtract \\ 2}} t - 2 \underbrace{\rightarrow}_{\substack{Divide \\ by\ 2}} \frac{t-2}{2}$$

Part B

$$b + 3$$
$$2b$$

Part C

$$t - 6$$
$$\frac{t}{2}$$

C. Reciprocals

Example 1.60

A saint has a magic pot that doubles gold coins that you put into it overnight. On Monday night, you put c gold coins into it.

A. How many coins will you have on Tuesday morning?
B. On Wednesday morning, you find d gold coins in the pot. How many gold coins were put into the pot on Tuesday night?
C. Find the relation between d and c?

Part A

On Monday night, we start with

$$c \text{ gold coins}$$

Overnight, the gold coins double, so on Tuesday morning

$$= 2c \text{ gold coins}$$

Part B

On Tuesday night, you must have put half of what you get Wednesday morning:

$$d \div 2 = d \times \frac{1}{2} = \frac{d}{2}$$

Note that when are we dividing by 2, it is the same as multiplying by the reciprocal of 2.

Part C

The number of coins on Tuesday night is the same, whether we write it in terms of d or in terms of c:

$$\frac{d}{2} = 2c \Rightarrow d = 4c$$

Example 1.61

The price of apples became one and a half from 6. What was the new price? It went back to the original price What was the original price?

$$6 \times \frac{3}{2} = 9$$
$$9 \div \frac{3}{2} = 9 \times \frac{2}{3} = 6$$

Example 1.62

When water freezes, its volume increases to eleventh-tenths of its liquid volume.

A. I start with w cubic cm of water, and freeze it. What is the volume of the ice?
B. I start with i cubic cm of ice, and melt it. What is the volume of the water?

The freezing of water results in increase in the volume of water:

$$Volume\ of\ ice = \frac{11}{10} \times \underbrace{w}_{Water}\ cm^3 = \frac{11}{10} w\ cm^3$$

Melting of ice results in reduction of volume.
We multiply by the reciprocal of $\frac{11}{10}$, which is $\frac{10}{11}$:

$$Volume\ of\ water = \underbrace{i}_{Ice}\ cm^3 \div \frac{11}{10} = i \times \frac{10}{11} = \frac{10}{11} i\ cm^3$$

Example 1.63

The dosage of medicine X is proportional to the weight of a person.

A. A person with 50 kg weight needs 5 gm of medicine. How much does a person who weighs 100 kg need?
B. Naina's mother takes a dose of x milligrams. Naina's mother's weight is double of her daughter's weight. What is the appropriate dose for Naina?
C. Naina's mother takes a dose of x milligrams. Naina's mother's weight is nine-fifths of her daughter's weight. What is the appropriate dose for Naina?

Part A

$$50\ kg \to 5\ gm$$
$$100\ kg \to 10\ gm$$

Part B

$$x \div 2 = \frac{x}{2}$$

Part C

Note that Naina's dosage must be less than her mother's dosage, since her weight is less than her mother's weight:

$$x \div \frac{9}{5} = x \times \frac{5}{9} = \frac{5x}{9}$$

Example 1.64

Maharaja Bhog has a *thali*, and gives a quantity of a grams of *aamras* to each diner. On every weekend day, there are c customers. The number of customers on a weekend is $\frac{5}{4}$ times the number of customers on a weekday. Find the quantity of *aamras* needed in a week.

On the weekend:

$$\underbrace{a}_{Aamras} \times \underbrace{c}_{Customers} = ac$$

Total Quantity on the weekend (Sat and Sun)

$$= 2 \times ac = 2ac$$

Weekend is more than weekday. Therefore, weekday is less than weekend.

$$ac \div \frac{5}{4} = ac \times \frac{4}{5} = \frac{4}{5}ac$$

Total quantity on weekdays (Mon to Fri, which is five days)

$$= 5 \times \frac{4}{5}ac = 4c$$

Total quantity in a week:

$$2ac + 4ac = 6ac$$

Example 1.65

10 apples cost Rs. 30. Find the cost of 15 apples.

We will use the unitary method.

$$10\ apples\ cost\ Rs.30$$

Find the cost of 1 apple by dividing both sides by 10:

$$\frac{10}{10}\ apples\ cost\ Rs.\frac{30}{10}$$

$$1 \ apples\ cost\ Rs. 3$$

Find the cost of 15 apples by multiplying both sides by 15:

$$1 \times 15\ apples\ cost\ Rs. 3 \times 15$$
$$15\ apples\ cost\ Rs. 45$$

Example 1.66

I can eat p pieces of pizza in m minutes. How many pieces of pizza can I eat in 1 minute?

A. How much time does it take to eat 1 piece of pizza?
B. If I enter a pizza eating contest, where I need to eat x pizzas, each with y pieces, how much time will it take?

$$m\ minutes: p\ pieces\ of\ pizza$$

Find the number of piece of pizza that can be eaten in one minute by dividing both sides by m:
Divide both side s by m:

$$\frac{m}{m}\ minutes: \frac{p}{m}\ pieces\ of\ pizza$$
$$1\ min: \frac{p}{m} pieces\ of\ pizza$$

The total number of pieces of pizza to eaten is

$$\underbrace{x}_{\substack{No.of \\ Pizzas}} \times \underbrace{y}_{\substack{No.of \\ Pieces\ per\ pizza}} = xy$$

Multiply both sides by xy

$$1 \times xy\ min: \frac{p}{m} \times xy\ pieces\ of\ pizza$$
$$xy\ min: \frac{p}{m} xy\ pieces\ of\ pizza$$

Multiply both sides by $\frac{m}{p}$

$$xy \times \frac{m}{p}\ min: \frac{p}{m} \times \frac{m}{p} xy\ pieces\ of\ pizza$$
$$xy\frac{m}{p}\ min: xy\ pieces\ of\ pizza$$

D. Rates

Example 1.67

3 cats can catch 6 rats in 1 minutes:

A. How much time does one cat take to catch one rat?
B. How many rats can a cat catch in 4 min?
C. How many cats are needed to catch 20 rats in a min?

Part A

$$3\ cats\ can\ catch\ 6\ rats\ in\ 1\ minutes$$

Since you have 3 cats, put them to work separately:

$$1st\ cat\ will\ catch\ 2\ rats$$

$$2nd\ cat\ will\ catch\ 2\ rats$$
$$3rd\ cat\ will\ catch\ 2\ rats$$

$$1\ cat\ can\ catch\ 2\ rats\ in\ 1\ min$$

Divide cats and rats by 3, and keep in mind that time remains unchanged:

$$1\ cat\ can\ catch\ 2\ rats\ in\ 1\ min$$

Part B

$$1\ cat\ can\ catch\ 2\ rats\ in\ 1\ min$$

$$Cat\ will\ catch\ 2\ rats\ in\ the\ 1st\ minute$$
$$Cat\ will\ catch\ 2\ rats\ in\ the\ 2nd\ minute$$
$$Cat\ will\ catch\ 2\ rats\ in\ the\ 3rd\ minute$$
$$Cat\ will\ catch\ 2\ rats\ in\ the\ 4th\ minute$$

$$1\ cat\ can\ catch\ 8\ rats\ in\ 4\ min$$

Part C

$$1\ cat\ can\ catch\ 2\ rats\ in\ 1\ min$$
$$2\ cats\ can\ catch\ 4\ rats\ in\ 1\ min$$
$$3\ cats\ can\ catch\ 6\ rats\ in\ 1\ min$$
$$\vdots$$
$$10\ cats\ can\ catch\ 20\ rats\ in\ 1\ min$$

Example 1.68

x cats can catch x rats in y minutes

A. How many rats can a cat catch in 1 min?
B. How many rats can a cat catch in x min?
C. How many rats will y cats catch in x minutes?

Part A

$$x\ cats\ can\ catch\ x\ rats\ in\ y\ minutes$$

Put each cat to work separately. In y minutes, each cat will catch 1 rat:

$$1\ cat\ can\ catch\ 1\ rat\ in\ y\ minutes$$

Since a cat takes y minutes to catch a rat, it can only catch:

$$1\ cat\ can\ catch\ \frac{1}{y}\ rats\ in\ 1\ minutes$$

Part B

In x minutes, the cat will catch x times of what it can catch in 1 min:

$$1\ cat\ can\ catch\ \frac{x}{y}\ rat\ in\ x\ minutes$$

Part V

$$y\ cats\ can\ catch\ x\ rats\ in\ x\ minutes$$

E. Multiple Variables

The questions till now made use of a single variable. In certain situations, more than one variable is necessary (or useful).

Example 1.69

Reena has r dollars and Saloni has s dollars more than Reena.

A. What is the money that Saloni has?
B. What is the total money that they have?
C. They pool their money together, and buy three games, each of which costs g dollars. What is the money total money left with them after they buy the games?

Part A

$$r + s$$

Part B

$$r + (r + s) = 2r + s$$

Part C

$$2r + s - 3g$$

Example 1.70

$$\textit{Read the information, and answer each question independently}$$

I bought a jar of chocolates. I ate c chocolates on Monday. I ate x more chocolates on Tuesday as compared to Monday. There were then five chocolates left in the jar, which my sister ate.

A. What was the number of chocolates in the jar when it was first bought?
B. If each chocolate costs 50 *cents*, what is the price of the jar of chocolates in dollars?
C. If the jar of chocolates cost d dollars, what is the price of one chocolate?

Part A

$$\begin{gathered} Monday = c \\ Tuesday = x + c \\ Sister = 5 \\ \underbrace{c}_{Monday} + \underbrace{x + c}_{Tuesday} + \underbrace{5}_{\substack{Remaining \\ Chocolates}} = 2c + x + 5 \end{gathered}$$

Part B

$$50\ cents = \frac{1}{2} dollars$$

$$\underbrace{(2c + x + 5)}_{No.of\ Chocolates} \times \underbrace{\frac{1}{2}}_{\substack{Cost\ per \\ Chocolates}} = \frac{1}{2}(2c + x + 5)$$

Part C

$$\frac{Total\ Cost}{No.of\ Chocolates} = \frac{d}{2c + x + 5}$$

Example 1.71

A restaurant earns d dollars for each weekday diner, and e dollars for each weekend diner. It costs the restaurant c dollars to make a dinner for each diner. In the first week of May 2022, the restaurant has 72 weekday diners and 112 weekend diners. Find the profit made by the restaurant.

$$\underbrace{72d}_{\substack{Revenue: Weekday \\ Diners}} + \underbrace{112e}_{\substack{Revenue: Weekend \\ Diner}} - \underbrace{(72+112)c}_{Cost}$$
$$= \underbrace{72d + 112e}_{Selling\ Price} - \underbrace{184c}_{Cost\ Price}$$

Example 1.72

The cost of 1 kg of steel is s dollars, while the cost of 1 kg of aluminium is a dollars. Find the cost of purchasing:

A. a kg of steel and s kg of aluminum
B. s kg of steel and a kg of aluminium

Part A

$$Cost\ of\ Steel = \underbrace{s}_{\substack{Price \\ of\ Steel}} \times \underbrace{a}_{\substack{Quantity \\ of\ steel}} = sa$$
$$Cost\ of\ Aluminium = \underbrace{a}_{\substack{Price \\ of\ Steel}} \times \underbrace{s}_{\substack{Quantity \\ of\ steel}} = as$$

$$Total = \underbrace{as}_{Steel} + \underbrace{sa}_{Aluminium} = as + as = 2as$$

Part B

$$(s)(s) + (a)(a) = s^2 + a^2$$

Example 1.73

A yellow candy costs c cents, and a green candy costs 3 cents more than a yellow candy. Find the cost of two yellow candies and three green candies.

$$2\ Yellow\ Candies = 2 \times c = 2c$$
$$3\ Green\ Candies = 3(c+3) = 3c + 9$$
$$Total\ Cost = \underbrace{2c}_{\substack{Yellow \\ Candies}} + 3c + 9 = 5c + 9$$

Example 1.74

A yellow candy costs c cents, and a green candy costs 3 cents more than a yellow candy. John and Jake get x cents and y *cents*, respectively for their allowance every weekday. They save the money for an entire week. John purchases three yellow candies and two green candies and Jake purchases two yellow candies and four green candies. Find the total money left with Jake and John

	Yellow Candy	Green Candy	Total
	c	$c+3$	
John	$3c$	$2c+6$	$5c+6$
Jake	$2c$	$4c+12$	$6c+12$
Total	$5c$	$6c+18$	

$$(5x + 5y) - (11c + 18)$$

Example 1.75

A yellow candy costs p cents, and a green candy costs 2 cents less than a yellow candy. John purchases two yellow candies and five green candies. Find the money left with John if every month gets x cents for his allowance, and he makes his purchase from his allowance for one week. (For this question, assume that a month has four weeks)

	Yellow Candy	Green Candy	Total
	p	$p-2$	
John	$2p$	$5p-10$	$7p-10$

$$\frac{x}{4}-(7p-10)=\frac{x}{4}-7p+10$$

Example 1.76: Finding Totals

How many video games do we own in all, if every member in my family owns:

A. 3 games, and there are 12 members in my family, including myself.
B. 3 games, and there are m members in my family, including myself.
C. x games, and there are 7 members in my family, including myself.
D. a games, and there are b members in my family, including myself.

$$Total\ Videogames=\underbrace{3}_{\substack{Videosgames\\ per\ Person}}\times\underbrace{12}_{No.of\ People}=36$$
$$Total\ Videogames=\underbrace{3}_{\substack{Videosgames\\ per\ Person}}\times\underbrace{m}_{No.of\ People}=3m$$
$$Total\ Videogames=\underbrace{x}_{\substack{Videosgames\\ per\ Person}}\times\underbrace{7}_{No.of\ People}=7x$$
$$ab$$

Example 1.77: Finding Per Person

How many video games does each person in my family own, if each person has an equal number of video games, and:

A. I have twelve video games in my family and four people in my family, including myself
B. I have sixteen video games in my family and there are p people in my family, including myself
C. I have sixteen video games in my family and there are p people in my family, not including myself.
D. I have x video games in my family, and there are 12 people in my family, not including myself

$$Videosgames\ per\ Person=\frac{Total\ Video\ Games}{No.of\ Family\ Members}$$

Part A

$$\frac{12}{4}=3$$

Part B

$$\frac{16}{p}$$

Part C

$$\frac{16}{p+1}$$

Part D

$$\frac{x}{13}$$

Example 1.78: Finding Changed Values

A. I have ten homework problems to do. I do two problems every day. After four days, how many problems do I still need to do?
B. I have 45 of my favorite chocolates currently. Every day, I eat three of them. After three days, how many chocolates am I left with?
C. I have h homework problems to do. I do two problems every day. After four days, how many problems do I still need to do?
D. I have c of my favorite chocolates currently. Every day, I eat three of them. After three days, how many chocolates am I left with?
E. I have h homework problems to do. I do t problems every day. After four days, how many problems do I still need to do?
F. I have c of my favorite chocolates currently. Every day, I eat d of them. After three days, how many chocolates am I left with?
G. I have h homework problems to do. I do t problems every day. After d days, how many problems do I still need to do?
H. I have c of my favorite chocolates currently. Every day, I eat y chocolates. After x days, how many chocolates am I left with?

Part A

$$10 - 2(4) = 10 - 8 = 2$$

Part B

$$45 - 3(3) = 45 - 9 = 36$$

Part C

$$h - 2(4) = h - 8$$

Part D

$$c - 3(3) = c - 9$$

Part E

$$h - 4t$$

Part F

$$c - 3d$$

Part G

$$h - td$$

Part H

$$c - xy$$

Example 1.79: Types of Integers

A. (*Even and Odd Integers*) Write an expression that represents all even integers. Write an expression that represents all odd integers.
B. (*Addition*) Harsh thinks of an odd integer. Myra thinks of an even integer. Kamal decides to add the two integers that Harsh and Myra thought of. Is the integer that Kamal gets even or odd? Repeat the question if both integers being added are odd. Repeat the question if both integers being added are even.
C. Represent the sum of

Part A

Even Number

Note that even numbers are all the multiples of 2. Hence, if n is an integer, then, any even number must be of the form:

$$2n$$

Odd Numbers

Note that any odd number is always one more than a multiple of 2.

$$5 = 4 + 1 = 2(2) + 1$$
$$13 = 12 + 1 = 2(6) + 1$$

$$Odd\ Integer = 2n+1, where\ n\ is\ an\ integer$$

Part B

$$\underbrace{2n+1}_{Odd} + \underbrace{2n}_{Even} = \underbrace{4n+1}_{Odd}$$

$$\underbrace{2n+1}_{Odd} + \underbrace{2n+1}_{Odd} = \underbrace{4n+2}_{Even}$$

$$\underbrace{2n}_{Even} + \underbrace{2n}_{Even} = \underbrace{4n}_{Even}$$

Example 1.80: Consecutive Integers

A. Amar, Akbar and Anthony think of three integers. Their father notes that the three integers are consecutive. Represent the three numbers using three different variables. Then, represent them using a single variable.
B. Amar, Akbar and Anthony think of three consecutive even integers. Represent the three numbers using using a single variable.
C. Amar, Akbar and Anthony think of three consecutive odd integers. Represent the three numbers using using a single variable.

Part A

We can assume three different variables, like this:

$$x, \quad y, \quad z\ with\ x < y < z$$

We can write using only one variable by using the information that the integers are consecutive:

$$x, \quad x+1, \quad x+2$$

In fact, in many problems, it becomes even more useful to consider the integers as:

$$y-1, \quad y, \quad y+1$$

Part B

You can represent the integers as:

$$2n-2, \quad 2n, \quad 2n+2$$

Or also as:

$$2n, \quad 2n+2, \quad 2n+4$$

Part C

We can let the integers be:

$$2n-1, \quad 2n+1, \quad 2n+3$$

Or also

$$2n+1, \quad 2n+3, \quad 2n+5$$

Example 1.81

Find an expression for the product of

A. Three consecutive integers
B. Three consecutive even integers

$$(a-1)(a)(a+1) = (a)(a^2-1) = a^3 - a$$

$$(2n-2)(2n)(2n+2) = 2(n-1) \times 2(n) \times 2(n+1) = 8(n^3 - n)$$

Example 1.82

Use a suitable expression that you found earlier to evaluate the following:

A. $2 \times 3 \times 4$

$$2 \times \underbrace{3}_{=a} \times 4 \Rightarrow a^3 - a = 3^3 - 3 = 27 - 3 = 24$$

Example 1.83: Perimeter

Find the perimeter in each case:

A. An equilateral triangle has all its sides equal. One side of an equilateral triangle has length s
B. A square is a quadrilateral with equal sides equal. One side of a square has length s.

C. A rectangle is a quadrilateral with two pairs of opposite sides with equal length. A rectangle has length l and breadth b.
D. A regular hexagon is a six-sided figure, with equal sides, and each side has length s.
E. A regular polygon has equal sides, and each side has length s.

Part A: Triangle

$$Side = s \Rightarrow Perimeter = 3 \times s = 3s$$

Part B: Square

$$Side = s \Rightarrow Perimeter = 4 \times s = 4s$$

Part C: Rectangle

$$length = l, breadth = b \Rightarrow Perimeter = p = l + l + b + b = 2l + 2b = 2(l + b)$$

Part D: Regular Hexagon

$$length\ of\ side = s, \text{No. of Sides} = 6 \Rightarrow Perimeter = p = 6s$$

Part E: Regular Polygon

$$length\ of\ side = \text{s}, \text{No. of Sides} = n \Rightarrow Perimeter = p = ns$$

1.84: Greek Letters

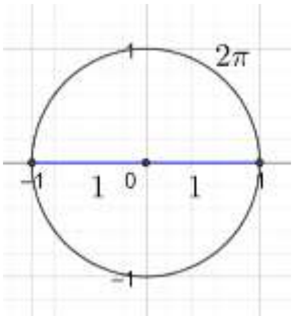
- π is a letter from the Greek language.
- π is the ratio between the diameter and the circumference of a circle.

- π is an irrational number
  - It cannot expressed as $\frac{p}{q}$, where p and q are integers.
  - It also cannot be expressed as a terminating decimal
- You cannot find the exact value of π.
- For calculations, we can use approximate values of π.

There are two very common approximations for π:

$$\pi \underbrace{\approx}_{\substack{approximately \\ equal\ to}} \frac{22}{7}$$

$$\pi \approx 3.14$$



Example 1.85

Choose the correct option

The value of π is:

A. $\frac{22}{7}$
B. 3.14
C. Both of the above
D. None of the above

Example 1.86

The diagram alongside shows a circle with radius 1, and diameter 2. Identify the circumference of the circle.

$$Circumference = D \times \pi = 2 \times \pi = 2\pi \underbrace{\approx}_{approx} 2(3.14) = 6.28$$

Example 1.87

A circle has perimeter (also called circumference) which is 2π times its radius. Find the circumference of a circle with radius r. (Answer in terms of π and r).

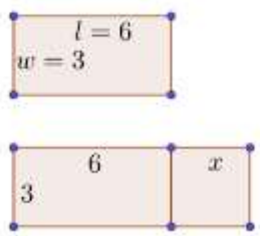
$$radius = r \Rightarrow Circumference = C = 2\pi r$$

Area

Example 1.88

A. Sally's dinner table is 6 feet long, and 3 feet wide. She has a large group of friends coming over for dinner, and they won't fit, so she makes the table longer by adding an extra piece that is x feet long. Find the area of the original table, and the table after she makes it longer.
B. A playground measures 30 feet by 100 feet long. A new road is built and y feet are removed from the length. Find the new area of the playground.
C. I need to buy 10 textbooks for my college. Each textbook costs d dollars. I purchase t textbooks from my money, and the rest are bought by my parents. How much money do my parents need to pay?
D. I bought P pizzas from a restaurant at a cost of d dollars each. Each pizza has p slices. I share the pizzas with my friends, and everyone pays for their share. If a friend of mine had x slices, how much should he pay?

Part A



Area of the original table

$$= 3 \times 6 = 18$$

After adding the extra piece, the

$$Length\ of\ Table = 6 + x, \qquad Width = 3$$

Area of the table after adding the extra piece:

$$= lw = (6 + x)(3) = 18 + 3x$$

Part B

$$30(100 - y) = 3000 - 30y$$

Part C

$$10d - td$$

Part D

$$Cost\ of\ 1\ pizza = d$$

$$Cost\ of\ 1\ slice\ of\ pizza = \frac{d}{p}$$

$$Cost\ of\ x\ slices\ of\ pizza = \frac{xd}{p}$$

Note:
$Number\ of\ pizzas = P$ is extra information.

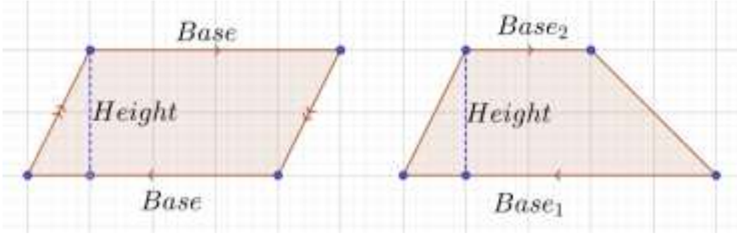
Example 1.89: Area Formulas

A. The area of a square is the length of its side multiplied by itself. Find the area of a square with side length s.
B. The area of a rectangle is the product of its length and width. Find the area of a rectangle with length l, and width w.
C. The area of a parallelogram is the product of its base and its height. Find the area of a parallelogram with base b, and height h.
D. The area of a rhombus is half the product of the lengths of its diagonals. Find the area of a rhombus with length of the first diagonal d_1 and length of the second diagonal d_2.
E. The area of a trapezoid is given by the product of its height, and the average length of its bases. Find the area of a trapezoid with height h, first base b_1 and second base b_2. Recall that the average of two numbers is given by half the sum of the two numbers.

$$Area\ of\ Square = Side \times Side = s \times s = s^2$$
$$Area\ of\ Rectangle = Length \times Width = lw$$
$$Area\ of\ Parallelogram = Base \times Height = bh$$
$$Area\ of\ Rhombus = \frac{Diagonal_1 \times Diagonal_2}{2} = \frac{d_1 d_2}{2}$$

$$Area\ of\ Trapezoid = Height\left(\frac{Base_1 + Base_2}{2}\right) = h\left(\frac{b_1 + b_2}{2}\right)$$

Example 1.90



- Remember to write the units when writing your answers for area.
- Units of area are always in squared units ($inches^2, feet^2, meter^2$)
- Answers without units are incomplete (and hence, incorrect).

A. A square has a side length of 3 inches. Find its area.
B. A rectangle has length of 6 inches, and width of 9 inches. Find its area.
C. A parallelogram has base $\frac{1}{2}$ meters and height $\frac{2}{3}$ meters. Find the area of the parallelogram.
D. A rhombus has length of diagonals 1.5 cm and 2.5 cm respectively. Find the area of the rhombus.
E. The trapezoid drawn alongside (diagram not drawn to scale) has length of smaller base 3 inches, length of longer base 6 inches and height 6 inches. Find the area of this trapezoid.
F. The trapezoid drawn alongside (diagram not drawn to scale) has length of smaller base 1.5 inches, length of longer base 2.5 and height 1.25 inches. Find the area of this trapezoid.

Part A

$$Area\ of\ Square = 3\ inches \times 3\ inches = 9\ inches^2$$

Part B

$$Area\ of\ Rectangle = 6\ inches \times 9\ inches = 54\ inches^2$$

Part C

$$Area\ of\ Parallelogram = \frac{1}{2}m \times \frac{2}{3}m = \frac{1}{3}m^2$$

Part D

$$Area\ of\ Rhombus = \frac{d_1 d_2}{2} = \frac{1.5 \times 2.5}{2} = \frac{\frac{3}{2} \times \frac{5}{2}}{2} = \frac{\frac{15}{4}}{2} = \frac{15}{4} \times \frac{1}{2} = \frac{15}{8}cm^2$$

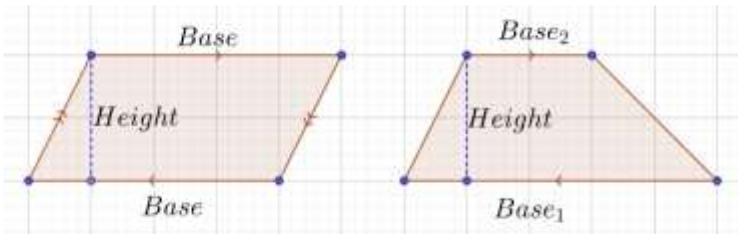
Part E

$$Area\ of\ Trapezoid = 6\left(\frac{3+6}{2}\right) = 6\left(\frac{9}{2}\right) = 3 \times 9 = 27\ inches^2$$

Part F

$$Area\ of\ Trapezoid = 1.25\left(\frac{1.5+2.5}{2}\right) = 1.25\left(\frac{4}{2}\right) = 2.5\ inches^2$$

Example 1.91



A. A square has a side length of 3 inches. Find its area in square feet.
B. A rectangle has length of 6 inches, and width of 9 inches. Find its area in square feet.
C. A parallelogram has base $3x$ and height $\frac{y}{9}$. Find the area of the parallelogram.
D. A rhombus has length of diagonals 3 cm and 5 cm respectively. Find the area of the rhombus.
E. A rhombus has length of diagonals $\frac{x}{2}$ and $4y$ cm respectively. Find the area of the rhombus.

F. The trapezoid drawn alongside (diagram not drawn to scale) has length of smaller base 4 inches, length of longer base 1 foot and height 6 inches. Find the area of this trapezoid.
G. The trapezoid drawn alongside (diagram not drawn to scale) has length of smaller base $\frac{x}{3}$ inches, length of longer base $\frac{x}{2}$ and height $5x$ inches. Find the area of this trapezoid.

Part A

$$s = 3\ inches = \frac{1}{4} feet$$
$$Area = s^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}\ ft^2$$

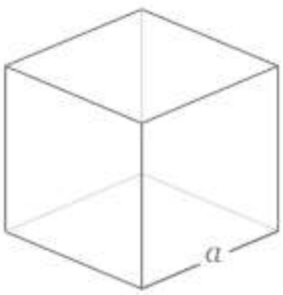
$$Area\ of\ Rhombus = \frac{d_1 d_2}{2} = \frac{3 \times 5}{2} = \frac{15}{2}$$

Conversion
First, note that not all measurements have the same units. So, convert everything to inches.
$$Length\ of\ longer\ base = 1\ foot = 12\ inches$$

Determine the Variables
$$h = 6\ inches$$
$$b_1 = 4\ inches$$
$$b_2 = 12\ inches$$

$$Area\ of\ Trapezoid = h\left(\frac{b_1 + b_2}{2}\right) = 6\left(\frac{4 + 12}{2}\right) = 6 \times 8 = 48\ inches^2$$

Three Dimensional Figures



Example 1.92: Cube

A. The volume of a cube is given by the length of the edge of its side, multiplied by itself three times. Find the volume of a cube with side length s.
B. A cube has six faces, each with equal area. Each face of a cube is a square. Find the surface area of a cube, with side length s.
C. A cube has side length 3. Find its volume. Find its surface area. Find the length of its edges.

Part A
$$Volume\ of\ Cube = s \times s \times s = s^3$$

Part B
Surface Area of One Face
$$= Area\ of\ Square = s \times s = s^2$$
Surface Area of Six Faces
$$= 6 \times s^2 = 6s^2$$

Part C
Since the side length of the cube is 3, we must have:

$$s = 3$$

$$Volume = s^3 = 3^3 = 3 \times 3 \times 3 = 27\ units^3$$
$$Surface\ Area = 6s^2 = 6 \times 3^2 = 6 \times 9 = 54\ units^2$$

Example 1.93

A. The volume of a cuboid is given by the product of its length, width and height. Find the volume of a cuboid with length l, width w, and height h.

B.

Example 1.94

A square has a length of side s. The side length is increased by 1 unit. Find the

A. Side length
B. perimeter of the new square.
C. area of the new square

$$Side\ Length = s + 1$$
$$Perimeter = 4(s + 1) = 4s + 4$$
$$Area = (s + 1)^2 = (s + 1)(s + 1) = s(s + 1) + 1(s + 1) = s^2 + s + s + 1 = s^2 + 2s + 1$$

Example 1.95

A rectangle has length l and width w. The length is increased by 1 unit, and the width is decreased by 1 unit. Find the new

A. Length
B. Width
C. Perimeter
D. Area

$$Length = l + 1$$
$$Width = w - 1$$
$$Perimeter = 2(l + 1) + 2(w - 1) = 2l + 2 + 2w - 2 = 2l + 2w$$
$$Area = (l + 1)(w - 1) = l(w - 1) + 1(w - 1) = lw - l + w - 1 = lw + w - l + 1$$

A. Triangles

Example 1.96

A. An isosceles triangle has base $2x + 2$ and one side $x + 2$. Find the perimeter.
B. The length of one side of an equilateral triangle is $\frac{1}{3}x + \frac{1}{9}$. Find the perimeter.

Part A

$$P = B + 2S = (2x + 2) + 2(x + 2) = 2x + 2 + 2x + 4 = 4x + 6$$

Part B

$$P = 3\left(\frac{1}{3}x + \frac{1}{9}\right) = x + \frac{1}{3}$$

B. Quadrilaterals

Example 1.97

A. The side length of a square is $x+3$ units. Find its perimeter and its area.
B. The length and breadth of a rectangle are $s+4$ and $s-2$, respectively. Find the area and the perimeter.

Part A

$$A = s^2 = (x+3)^2 = x^2 + 6x + 9$$

Part B

$$A = lb = (s+4)(s-2) = s^2 + 4s - 2s - 8 = s^2 + 2s - 8$$
$$P = 2(l+b) = 2[(s+4)+(s-2)] = 2[2s+2] = 4s+4$$

1.98: Area of a Trapezoid

$$\boldsymbol{A = \frac{h(b_1+b_2)}{2}, h = height, b_1 = first\ base, b_2 = second\ base}$$

Example 1.99

Find the area of a trapezoid with longer base $x+4$, shorter base $x-3$, and height $x+2$.

$$A = \frac{h(b_1+b_2)}{2} = \frac{(x+2)[(x+4)+(x-3)]}{2} = \frac{(x+2)(2x+1)}{2} = \frac{2x^2+5x+2}{2}$$

1.100: Perimeter of a Rhombus

$$\boldsymbol{P = 4s, s = side\ length}$$

Recall that all four sides of a rhombus have equal length

Example 1.101

Find, in terms of x, the perimeter of a rhombus, with side length s, and one-third of s is $\left(\frac{1}{4}x+\frac{1}{5}\right)$

$$s = 3\left(\frac{1}{4}x+\frac{1}{5}\right) = \frac{3}{4}x + \frac{3}{5} \Rightarrow P = 4s = 4\left(\frac{3}{4}x+\frac{3}{5}\right) = 3x + \frac{12}{5}$$

1.102: Area of a Kite

$$\boldsymbol{A = \frac{d_1 d_2}{2}, d_1 = first\ diagonal, d_2 = second\ diagonal}$$

Example 1.103

A kite has one diagonal of length $l+\frac{1}{3}$, and a second diagonal of length $m-\frac{1}{2}$. Find its area.

$$A = \frac{d_1 d_2}{2} = \frac{\left(l+\frac{1}{3}\right)\left(m-\frac{1}{2}\right)}{2} = \frac{lm - \frac{1}{2}l + \frac{1}{3}m - \frac{1}{6}}{2} = \frac{lm}{2} - \frac{1}{4}l + \frac{1}{6}m - \frac{1}{12}$$

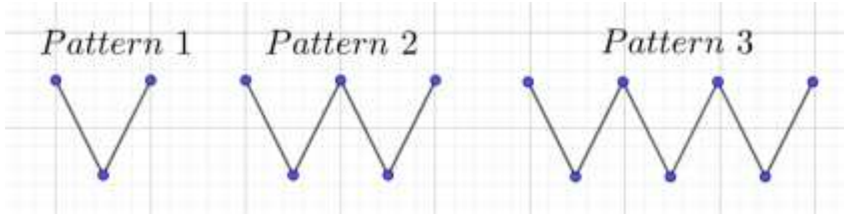
1.4 Patterns and Sequences

A. Pattern Basics

Example 1.104

The pattern on the left is continued.

A. List the number of matchsticks in the first few patterns. Find the number of matchsticks in the tenth pattern? n^{th} pattern?
B. List the number of blue dots in the first few patterns. Find the number of matchsticks in the tenth pattern? n^{th} pattern?



Part A

$$2,4,6,8,10,\ldots$$
$$10^{th} Pattern = 20\ Matchsticks$$
$$n^{th}\ pattern = 2n\ matchsticks$$

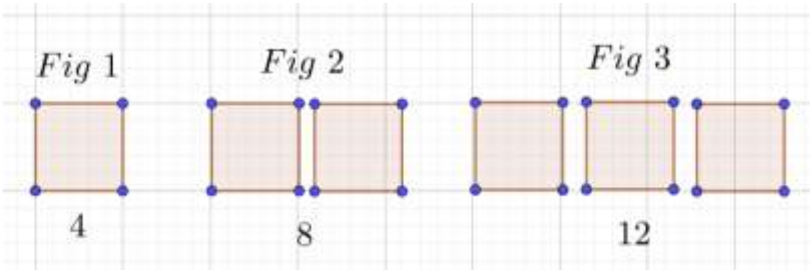
Part B

$$List: 3,5,7,\ldots$$
$$10th\ Pattern: 21\ Blue\ Dots$$
$$n^{th}\ pattern = (2n+1)\ matchsticks$$

Example 1.105

What is the number of matchsticks:

A. needed for the first few figures?
B. number of matchsticks needed for the 5th figure? For the n^{th} figure?



Part A

$No. of\ Matchsticks$

$$= \underbrace{4}_{\substack{1st\\Figure}}, \underbrace{8}_{\substack{2nd\\Figure}}, \underbrace{12}_{\substack{3rd\\Figure}}$$

Part B

Note that each figure adds one square, and each square has four matchsticks.

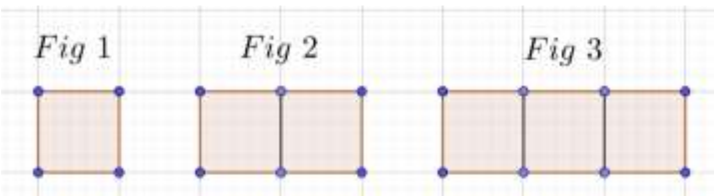
$$5th\ Figure = 5 \times 4 = 20\ Matchsticks$$

In the n^{th} figure, there will be n squares, and the number of matchsticks

$$= 4 \times n = 4n$$

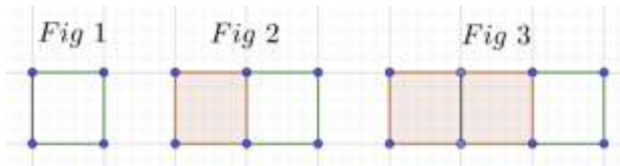
Example 1.106

A. What is the number of matchsticks needed for the first few figures?
B. The pattern continues. Find the number of matchsticks in the 5th figure? the n^{th} figure?



Part A

$$\underbrace{4}_{\substack{1st\\Figure}}, \underbrace{7}_{\substack{2nd\\Figure}}, \underbrace{10}_{3rd\ Figure}$$

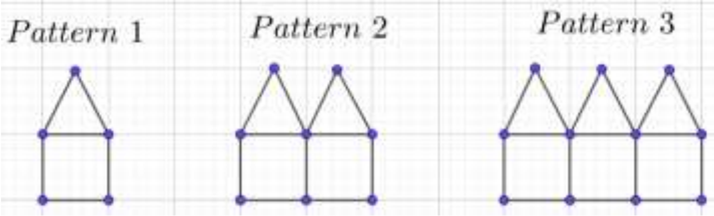


Part B

Note that every time we move from one figure to the next, we add three matchsticks. However, the first figure has one additional matchstick (four), as compared to three.

Hence, the number of matchsticks needed for the n^{th} figure is:

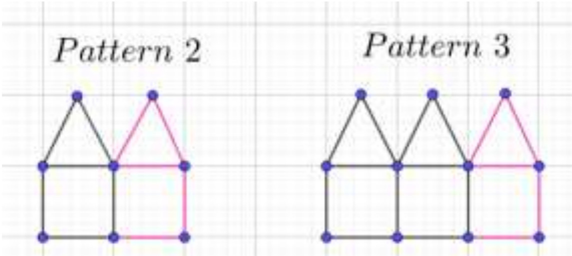
$$3n + 1$$



Example 1.107

A. What is the number of matchsticks needed for the first few figures?

B. The pattern continues. Find the number of matchsticks in the 5th figure? the n^{th} figure?



The first house needs

$$6\ Matchsticks$$

Every additional house needs five more matchsticks:

$$6, 6 + 5, 6 + 10, \ldots$$

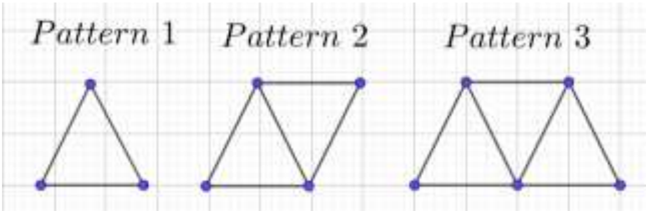
$$6, 6 + 5, 6 + 10, \ldots$$

The final count of matchsticks is:

$$6, 11, 16, \ldots = 5 + 1, 10 + 1, 15 + 1, \ldots$$

Which can be generalized to give the number of matchsticks in the n^{th} house as:

$$5n + 1$$



Example 1.108

A. What is the number of matchsticks needed for the first few figures?

B. The pattern continues. Find the number of matchsticks in the 5th figure? the n^{th} figure?

$$2n + 1$$

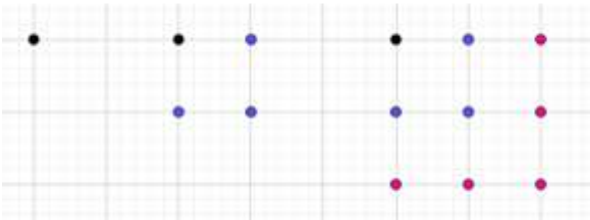
B. More Patterns

Example 1.109: Square Numbers

In the first round of a game, Manish draws a black dot. In the second round, he draws three blue dots. In the third round, he draws five pink dots.

A. Find the number of dots drawn in the fifth round. The n^{th} round.

B. Find the number of dots drawn upto and including in the fifth round. The n^{th} round.



Part A

$$1, 3, 5, \ldots$$

$$5th\ Round: 9$$

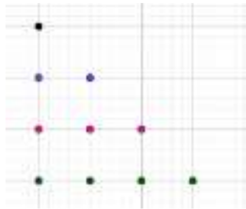
$$n^{th}\ Round = 2n - 1$$

Part B

$$Total\ Dots: 1, 4, 9, 16, 25, \ldots = 1^2, 2^2, 3^2, 4^2, 5^2, \ldots$$

$$5^{th} Round: 25$$

$$n^{th} Round: n^2$$

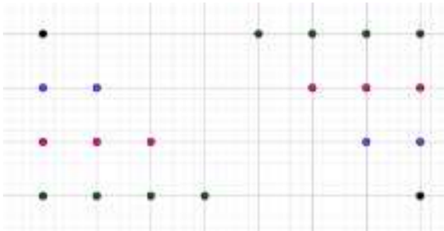


Example 1.110: Triangular Numbers

Mohan draws dots in the pattern shown alongside. He puts a single dot in the first row, two dots in the second row, three dots in the third row, and so on.

A. Find the number of dots in the n^{th} row.
B. Find the total of dots upto and including the n^{th} row.

Pattern with 4 rows

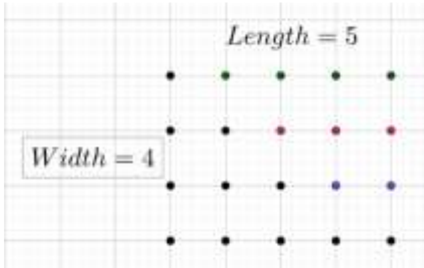


Step I: Draw the same pattern *upside down*.
Step II: Bring the two patterns close together, and notice that they form a rectangle with width 4 and length 5, and hence the total number of dots in the rectangle is:

$$4 \times 5 = 20$$

Step III: Remember the rectangle has double the number of dots in the original pattern, and hence, the final answer is:

$$\frac{4 \times 5}{2} = 2 \times 5 = 10$$



Pattern with n rows

If you repeat the same steps, in a pattern with n rows, the rectangle obtained has

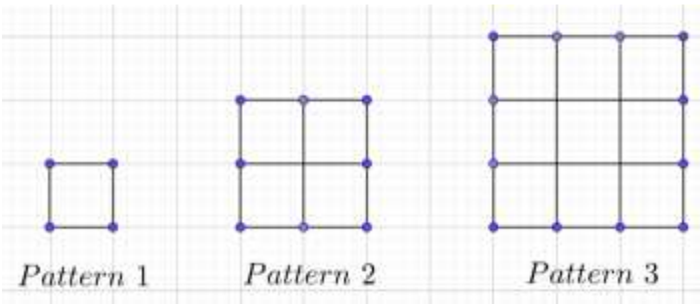
$$\underbrace{n}_{Width} \times \underbrace{(n+1)}_{Length} = n(n+1)$$

And since the rectangle has double the number of dots that we want, the actual number of dots is:

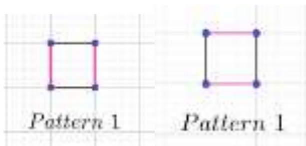
$$\frac{n(n+1)}{2}$$

Example 1.111

A. Find the number of matchsticks in the first few patterns.



Pattern 1



$$Column\ Total = \underbrace{2}_{\substack{No.\ of \\ Columns}} \times \underbrace{1}_{\substack{MS\ per \\ Column}} = 2$$

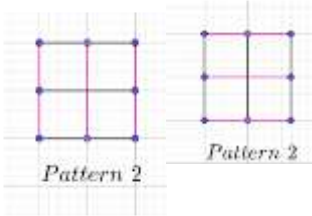
$$Row\ Total = \underbrace{2}_{\substack{No.\ of \\ Rows}} \times \underbrace{1}_{\substack{MS\ per \\ Row}} = 2$$

$$Final\ Answer = 2 + 2 = 4$$

Pattern 2

$$Column\ Total = \underbrace{3}_{\substack{No.\ of \\ Columns}} \times \underbrace{2}_{\substack{MS\ per \\ Column}} = 6$$

$$Row\ Total = \underbrace{3}_{\substack{No.\ of \\ Rows}} \times \underbrace{2}_{\substack{MS\ per \\ Row}} = 6$$



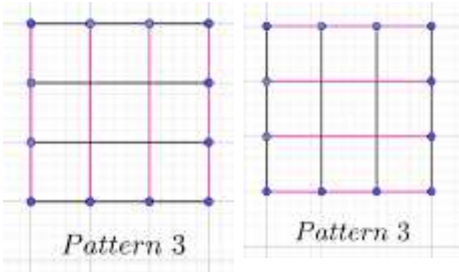
$$Final\ Answer = 6 + 6 = 12$$

Pattern 3

$$Column\ Total = \underbrace{4}_{\substack{No.\ of \\ Columns}} \times \underbrace{3}_{\substack{MS\ per \\ Column}} = 12$$

$$Row\ Total = \underbrace{4}_{\substack{No.\ of \\ Rows}} \times \underbrace{3}_{\substack{MS\ per \\ Row}} = 12$$

$$Final\ Answer = 12 + 12 = 24$$



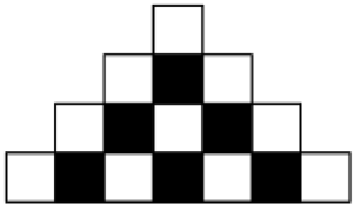
Pattern n

$$Column\ Total = \underbrace{n+1}_{\substack{No.\ of \\ Columns}} \times \underbrace{n}_{\substack{MS\ per \\ Column}} = n(n+1)$$

$$Row\ Total = \underbrace{n+1}_{\substack{No.\ of \\ Rows}} \times \underbrace{n}_{\substack{MS\ per \\ Row}} = n(n+1)$$

$$Final\ Answer = n(n+1) + n(n+1) = 2n(n+1)$$

Example 1.112



A. A "stair-step" figure is made of alternating black and white squares in each row. Rows 1 through 4 are shown. All rows begin and end with a white square. The number of black squares in the 37th row is: (AMC 8 1985/7)

B. The number of black squares in the n^{th} row is:

C. The number of white squares in the n^{th} row is:

D. The total number of squares in the n^{th} row is:

Part A

Row	1	2	3	4		37
No. of Black Squares	0	1	2	3		36

Part B

Note that the number of black squares is one less than the row number. Hence, the number of black squares in the n^{th} row is:

$$n - 1\ squares$$

Part C

Row	1	2	3	4	37	n
No. of White Squares	1	2	3	4	37	n

$$n\ squares$$

Part D

$$\underbrace{n-1}_{\substack{Black\\Squares}} + \underbrace{n}_{\substack{White\\Squares}} = 2n-1$$

Row	1	2	3	4		n
No. of White Squares	1	3	5	7		$2n-1$

C. Sequences

Example 1.113

Write an expression for the n^{th} term of the following sequences:

A. $1,2,3,4,5,\dots$
B. $4,5,6,7,8,\dots$
C. $2,4,6,8,\dots$
D. $5,10,15,20,\dots$
E. $-1,-2,-3,-4,\dots$
F. $-3,-6,-9,\dots$
G. $5,4,3,2,1,0,-1,\dots$

Part A

$$t_n = n$$

Part B

$$t_n = n+3$$

Part C

The numbers are all multiple of 2. Hence:

$$t_n = 2n$$

Part D

The numbers are all multiple of 5. Hence:

$$t_n = 5n$$

Part E

$$t_n = -n$$

Part F

$$t_n = -3n$$

Part G

Note if we add 6 to every term in Part E, we get Part G. Hence

$$t_n = -n+6 = 6-n$$

Example 1.114: Arithmetic Sequences

Write an expression for the n^{th} term of the following sequences:

A. $4,7,10,13,\dots$
B. $7,12,17,22,\dots$
C. $3,7,11,15,19,\dots$
D. $12,22,32,42,\dots$
E. $99,199,299,399,\dots$
F. $3,0,-3,-6,-9,\dots$

Part A

The numbers are not multiples of any natural number. However, note that:

$$7-4=3$$
$$10-7=3$$
$$13-10=3$$

In other words, two consecutive terms have a difference of three.

Suppose we try:

$$t_n = 3n \Rightarrow Sequence = \{3,6,9,12,\dots\}$$

This does not match the pattern, but every number is exactly one less than the number that we want.

Hence, the final answer is:

$$t_n = 3n+1 \Rightarrow Sequence = \{4,7,10,13,\dots\}$$

Part B

Note that each number is two more than a multiple of 5:

$$5n+2$$

Part C

Note that each number is one less than a multiple of four:

$$4n - 1$$

Part D

Note that each number is two more than a multiple of ten:

$$10n + 2$$

Part E

Note that each number is one less than a multiple of hundred:

$$100n - 1$$

Part F

The difference between two consecutive terms is

$$-3$$

However:

$$t_n = -3n \Rightarrow \{-3, -6, -9, \dots\}$$

Add 6 to each term, to get us the sequence that we want:

$$t_n = 6 - 3n$$

Example 1.115: Finding a particular term

In each question below, find the general term of the sequence. Use it to find the:

A. twenty fifth term of the sequence 3,4,5,6,7, ….
B. twelfth term of the sequence 6,12,18, ….
C. tenth term of the sequence $-2, -4, -6, \dots$.
D. Hundredth term of 4,9,14,19, …
E. Fiftieth term of 5,4,3,2, …
F. Twentieth term of 8,5,2, …

Part A

$$t_n = n + 2$$
$$t_{25} = 25 + 2 = 27$$

Part B

$$t_n = 6n$$
$$t_{12} = 12 \times 6 = 72/$$

Part C

$$t_n = -2n$$
$$t_{10} = -20$$

Part D

$$t_n = 5n - 1$$
$$t_{100} = 5(100) - 1 = 499$$

Part E

$$t_n = 6 - n$$
$$t_{50} = 6 - 50 = -44$$

Part F

$$t_n = 11 - 3n$$
$$t_{20} = 11 - 3(20) = 11 - 60 = -49$$

Example 1.116: Powers

A. 2,4,8,16, …
B. 1,3,7,15,31

Part A

$$t_1 = 2 = 2^1$$
$$t_2 = 4 = 2 \times 2 = 2^2$$
$$t_3 = 8 = 2 \times 2 \times 2 = 2^3$$
$$t_4 = 16 = 2 \times 2 \times 2 \times 2 = 2^4$$

$$t_n = 2^n$$

Part B

Note the connection between Part A and Part B. If you subtract one from each term in Part A, you get the sequence of Part B.

$$t_1 = 2 - 1 = 1$$
$$t_2 = 4 - 1 = 3$$
$$t_3 = 8 - 1 = 7$$
$$t_4 = 16 - 1 = 15$$
$$t_5 = 32 - 1 = 31$$

$$t_n = 2^n - 1$$

Example 1.117: Word Problems

A. A saint has a magic pot. It doubles anything left in it overnight. On Monday night, the saint put a mango into the empty pot. He did the same on Tuesday night and Wednesday night. He did not remove anything till Thursday morning. How many mangoes did he get on Thursday?

B. On Monday Morning, Evelyn has 4 problems left from her homework today. Her teacher gives her seven problems for homework every day from Monday to Friday, but she only manages to solve five problems every day. How many problems will she have to solve over the weekend to complete her homework?

C. A bacteria species doubles its population every hour in ideal conditions. A lab has a petri dish with ideal conditions for the bacteria to grow. The lab assistant comes in 10 am, and notes that the petri dish is completely filled with bacteria. Assuming that the bacteria just filled the dish, at what time was the dish $\frac{1}{4}th$ filled?

D. Jane started a rumor in school by telling two of her friends on Monday. Each of the friends she told the rumor to, told two friends the next day, and those friends did the same. If this pattern continues, how many people in the school, including Jane, will have heard of the rumor by Friday?

Part A

	Mon	Tue	Wed	Thu
Morning	0	2	6	14
Night	1	3	7	

Part B

	Mon	Tue	Wed	Thu	Fri
Morn	4				
Night	6	8	10	12	14

Part C

$$10\ am: Full$$
$$9\ am: Half\ Full$$
$$8\ am: One-fourth\ full$$

Part D

Leave aside Jane for the time being:

$$Monday: 2+1=3$$

On Tuesday, those two people told two people each, so four new people came to know about the rumor.

$$3+4=7$$

On Wednesday, the four new people told two people each, for a total of eight new people:

$$7+8=15$$

On Thursday, the eight new people told two people each, for a total of sixteen new people:

$$15+16=31$$

On Friday, the sixteen new people told two people each, for a total of thirty-two new people:

$$31+32=63$$

1.5 Simplifying Expressions

A. Adding Variables of the same type

1.118: Like Terms

Terms which have the exact same variables are like terms. Like terms are very important to recognize since you can add like terms together

Example 1.119

A. $x+x$
B. $y+y$
C. $a+a+a$
D. $b+b+b+b$
E. $q+q+q$
F. $r+r+p+p+p$

$$x+x=x\times 2=2x$$
$$y+y=y\times 2=2y$$
$$a+a+a=a\times 3=3a$$

$$b + b + b + b = b \times 4 = 4b$$
$$q + q + q = q \times 3 = 3q$$
$$r + r + p + p + p = r \times 2 + p \times 3 = 2r + 3p$$

Example 1.120

Adding Variables

A. $4x + 2x$
B. $2y + 3y$
C. $5x - 2x$
D. $4x - 6x$

$$4x + 2x = 6x$$
$$2y + 3y = 5y$$
$$5x - 2x = 3x$$
$$4x - 6x = -2x$$

1.121: Unlike Terms

Terms which do not have the exact same variable are called unlike terms. You cannot add unlike terms.

Example 1.122

Simplify each part below, if possible.

A. $4 + 4y$
B. $2 - 3y$

It cannot be simplified further.

Example 1.123

On simplifying $6 + 3y$, we get:

A. 9
B. $9y$
C. $9x$
D. None of the above

Since 6 and $3y$ are unlike terms, we cannot simplify them.

Example 1.124

Simplify by adding the like terms:

A. $4r + 4 + 3r + 2$
B. $5s - 3 - 3s + 9$
C. $5z - 3z - 3 + 9$
D. $5w + 3 + 9 - 6w$

$$4r + 3r + 4 + 2 = 7r + 6$$
$$5s - 3s - 3 + 9 = 2s + 6$$
$$2z + 6$$
$$5w - 6w + 3 + 9 = 12 - w$$

Word Problems

Example 1.125: Addition

A. Add $2a + 4$ to twice of $3a + 6$
B. Add $4a - 1$ to three times of $a + 2$

$$2a + 4 + 6a + 12 = 8a + 16$$
$$4a - 1 + 3a + 6 = 7a + 5$$

Example 1.126: Subtraction Basics

A. Subtract $4x$ from $3x$
B. Subtract $-4x$ from $3x$
C. Subtract $2x$ from $-5x$
D. Subtract $-2x$ from $-5x$

$$3x - 4x = -x$$
$$3x - (-4x) = 3x + 4x = 7x$$
$$-5x - 2x = -7x$$
$$-5x - (-2x) = -5x + 2x = -3x$$

Example 1.127: Subtraction

A. Subtract $4x - 3$ from $2x + 4$

$$2x + 4 - 4x + 3 = -2x + 7$$

1.128: Multiplication versus Numbers

Case I: Default

Supppse $a = 2$

$$0.1a = 0.1 \times a = 0.1 \times 2 = 0.2$$

Case II: When working with Numbers

A decimal number less than one has the digit 2 in the tenth's place. And it has two even, non-zero digits to the right of the decimal point. Identify all possible values of the number

The number is of the form

$$0.2a$$

- a is a digit from 0 to 9.
- $0.2a$ is a two digit number

$$0.2a \textit{ could be } \{0.22, 0.24, 0.26, 0.28\}$$

The second case is only applicable when the two bullet points are mentioned explicitly.

Example 1.129: Decimals

A. $0.3x + 0.2 + 0.5x + 0.7$
B. $0.4r - 0.5 + 0.2r - 0.9$
C. $-0.1s + 0.8 + 0.5s - 0.9$
D. $-0.3t - 0.2 - 0.5t - 0.7$
E. $+0.3a + 0.2 - 0.4a - 0.5$

$$0.8x + 0.9$$
$$0.6r - 1.4$$
$$0.4s - 0.1$$
$$-0.8t - 0.9$$
$$-0.1a - 0.3$$

Example 1.130: Fractions and Decimals

A. $\frac{1}{2}x + \frac{1}{3}x$
B. $\frac{1}{2}x - \frac{1}{3}x$
C. $\frac{1}{2}x + 0.3x$
D. $x - 0.01x$

$$\frac{1}{2}x + \frac{1}{3}x = \frac{3}{6}x + \frac{2}{6}x = \frac{5}{6}x$$
$$\frac{1}{2}x - \frac{1}{3}x = \frac{3}{6}x - \frac{2}{6}x = \frac{1}{6}x$$
$$\frac{1}{2}x + 0.3x = \frac{1}{2}x + \frac{3}{10}x = \frac{5}{10}x + \frac{3}{10}x = \frac{8}{10}x = \frac{4}{5}x$$
$$x - 0.01x = 0.99x$$

Example 1.131: Fractions in the Numbers

A. $3x + \frac{1}{2} + 4x + \frac{1}{3}$
B. $\frac{1}{2}x + 3 + \frac{2}{5}x + 5\frac{1}{2}$
C. $2x - \frac{1}{2} - 7x - \frac{1}{3}$
D. $\frac{2}{3}x + 4 - \frac{7}{2}x - 2$
E. $\frac{1}{5}x - \frac{2}{3} - \frac{5}{2}x - \frac{2}{5}$

Example 1.132: Longer Expressions

A. $2 + 3x + 4y + 2x + 5y + 8$
B. $3r - 2z + 4 - 4r + 3z + 2$
C. $\frac{3}{2}r - \frac{2}{3}z + \frac{4}{5} - \frac{4}{3}r + \frac{3}{2}z + \frac{2}{7}$
D. $0.3x + 0.4y + \frac{2}{3}x - \frac{3}{4}y + 0.6 + \frac{4}{7}$
E. $\frac{1}{2} + \frac{1}{3}x + \frac{1}{5}y - \frac{1}{4}x - \frac{1}{6}y$

$$4x + 2x = 6x$$
$$2y + 3y = 5y$$
$$5x - 2x = 3x$$
$$4x - 6x = -2x$$

$$4 + 4y = 4 + 4y$$
$$2 - 3y = 2 - 3y$$

$$\frac{1}{2}x + \frac{1}{3}x = \frac{5}{6}x$$
$$0.2x + 0.01x = 0.21x$$

$$\frac{1}{2}x + 0.3x = \frac{8}{10}x = 0.8x$$

$$\frac{4}{6}x - \frac{21}{6}x + 4 - 2 = -\frac{17}{6}x + 2$$
$$\frac{2}{10}x - \frac{25}{10}x - \frac{10}{15} - \frac{6}{15} = -\frac{23}{10}x - \frac{16}{15}$$

$$2 + 3x + 4y + 2x + 5y + 8 = 10 + 5x + 9y$$
$$-r + z + 6$$
$$\frac{9}{6}r - \frac{8}{6}r - \frac{4}{6}z + \frac{9}{6}z + \frac{28}{35} + \frac{10}{35} = \frac{1}{6}r + \frac{5}{6}z + \frac{38}{35}$$

Part D

In this question for each of x, y and the number, one of the terms is in decimals, and the other is fractions. Hence, we need to convert either the fractions to decimals, or vice versa.

In this case, converting to fractions is better and easier.

$$\frac{3}{10}x + \frac{2}{3}x + \frac{4}{10}y - \frac{3}{4}y + \frac{6}{10} + \frac{4}{7}$$
$$= \frac{9+20}{30}x + \frac{8-15}{20}y + \frac{42+40}{70}$$
$$= \frac{29}{30}x - \frac{7}{20}y + \frac{82}{70}$$
$$= \frac{29}{30}x - \frac{7}{20}y + \frac{41}{35}$$

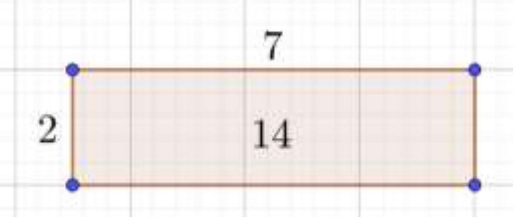
Part E

$$\frac{1}{2} + \frac{1}{3}x + \frac{1}{5}y - \frac{1}{4}x - \frac{1}{6}y = \frac{1}{2} + \frac{1}{12}x + \frac{1}{30}y$$

1.6 Distributive Property

A. Distributive Property

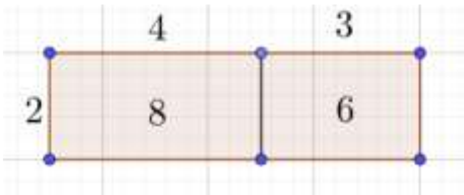
1.133: Distributive Property with Numbers



The area of the top rectangle is:

$$\underbrace{7}_{Length} \times \underbrace{2}_{Width} = 14$$

If we split the rectangle into two rectangles with the same width, but different lengths of 4 units, and 3 units, we get the diagram alongside.



This rectangle has area

$$2 \times 4 + 2 \times 3 = 8 + 6 = 14$$

However, we can also write:

$$2 \times 4 + 2 \times 3 = 2(4 + 3) = 2(7) = 14$$

Example 1.134

$$4 \times 18 = 4(10 + 8) = 4 \times 10 + 4 \times 8 = 40 + 32 = 72$$

Example 1.135

Suppose I wish to evaluate

$$2(3 + 5) = 2 \times 8 = 16$$

The distributive property for numbers is (for example):

$$2(3 + 5) = 2 \times 3 + 2 \times 5 = 6 + 10 = 16$$

Example 1.136

The 3 is next to the bracket. There is no sign between the two. That means it is multiplication. (Remember that multiplication takes higher priority over addition and subtraction).

$$3\left(\frac{1}{2} + \frac{1}{5}\right)$$

To simplify the expression, we could add the fractions. But there is another way – we can carry out the multiplication first.

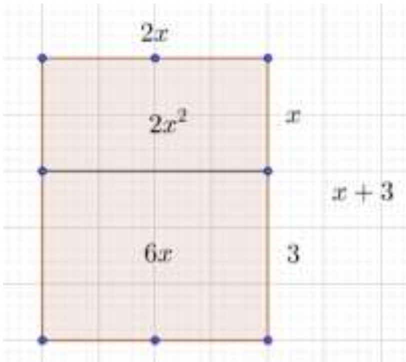
However, to do the multiplication, we need to multiply 3 with each term in the bracket:

$$3 \times \frac{1}{2} + 3 \times \frac{1}{5} = \frac{3}{2} + \frac{3}{5} = \frac{15}{10} + \frac{6}{10} = \frac{21}{10} = 2\frac{1}{10}$$

1.137: Distributive Property with Variables

$$2x(x + 3) = 2x(x) + 2x(3) = 2x^2 + 6x$$

$$4(\boldsymbol{x} + \boldsymbol{2y}) = \boldsymbol{4} \times \boldsymbol{x} + \boldsymbol{4} \times \boldsymbol{2y} = \boldsymbol{4x} + \boldsymbol{8y}$$



1.138: Distributive Property (Formula)

$$a(b + c) = a \times b + a \times c = ab + ac$$

Example 1.139

A. $3(x + y)$
B. $5(x - y)$
C. $2(3x + 4y)$
D. $9(-2x + 3y)$

$$3x + 3y$$
$$5(x - y) = 5x - 5y$$
$$2(3x + 4y) = 6x + 8y$$
$$9(-2x + 3y) = -18x + 27y$$

Example 1.140

A. $2(x + 3)$
B. $2(x - 3)$
C. $2(3 - x)$

$$2(x + 3) = 2x + 6$$
$$2(x - 3) = 2x - 6$$
$$2(3 - x) = 6 - 2x$$

Example 1.141

A. $\frac{1}{2}(2x + 8y)$
B. $\frac{1}{3}(12x - 15y)$

$$\frac{1}{2}(2x + 8y) = x + 4y$$
$$\frac{1}{3}(12x - 15y) = 4x - 5y$$

Example 1.142

A. $\frac{1}{2}(5x - 3y)$
B. $\frac{4}{7}(14x - 3y)$

$$\frac{1}{2}(5x - 3y) = \frac{5}{2}x - \frac{3}{2}y$$
$$\frac{4}{7}(14x - 3y) = 8x - \frac{12}{7}y$$

1.143: Distributing a minus sign

When a number outside a bracket (or parentheses) has a minus sign, the minus also distributes to each term in the bracket.

Example 1.144

A. $-2(x + y)$
B. $-2(x - y)$
C. $-2(-x + y)$
D. $-2(-x - y)$

$$-2(x + y) = -2x - 2y$$
$$-2(x - y) = -2x + 2y$$
$$-2(-x + y) = 2x - 2y$$
$$-2(-x - y) = 2x + 2y$$

1.145: Expressions

Expressions can be handled using the distributive property.

$$BODMAS$$

- $Brackets,$
- Of
- $Division, Multiplication$
- $Addition, Subtraction$

Example 1.146

A. $7-2(x+1)$
B. $9-3(y+2)$
C. $4-2(x-1)$
D. $6-3(x-y)$
E. $4-1(2-z)$

$$7-2\times(x+1)=7-2x-2=5-2x$$
$$9-3(y+2)=9-3y-6=3-3y$$
$$4-2(x-1)=4-2x+2=6-2x$$
$$6-3(x-y)=6-3x+3y$$
$$4-1(2-z)=4-2+z$$

Example 1.147

A. $\frac{1}{2}-\frac{1}{3}\left(x+\frac{1}{2}\right)$

$$\frac{1}{2}-\frac{1}{3}\left(x+\frac{1}{2}\right)=\frac{1}{2}-\frac{1}{3}x-\frac{1}{6}=\frac{1}{3}-\frac{1}{3}x$$

Example 1.148: Find the Mistake

$$\frac{2}{3}-\frac{2}{3}\left(x+\frac{1}{2}\right)=0\left(x+\frac{1}{2}\right)=0$$

$$\frac{2}{3}-\frac{2}{3}\left(x+\frac{1}{2}\right)=\frac{2}{3}-\frac{2x}{3}-\frac{1}{3}=\frac{2-2x-1}{3}=\frac{1-2x}{3}$$

Example 1.149

$$9-2+3=9-5=4$$

$$9-\underbrace{2+3}_{Wrong}=9-5=4$$

The mistake sign applies to the two and not to the 3. The correct way of doing it would be:

$$9+\underbrace{-2+3}_{Wrong}=9+1=10$$

Else, you can carry out the operations from left to right:

$$9-2+3=7+3=10$$

Example 1.150
Rishabh has three blueberry muffins. Each blueberry muffin costs two dollars. Radha has four mango muffins. The mango muffins also cost two dollars.

A. Find the total cost of the muffins by adding the cost of the blueberry muffins to the cost of the mango muffins.
B. Find the total cost of muffins by multiplying the total number of muffins with the cost per muffin.
C. Do both Part A and Part B give the same answer?

Part A

$$\underbrace{2\times 3}_{\substack{Blueberry\\ Muffins}} + \underbrace{2\times 4}_{\substack{Mango\\ Muffins}} = 6+8=14$$

Part B: Second Method

$$(3+4)2 = \underbrace{7}_{\substack{No.of\\ Muffins}} \times \underbrace{2}_{\substack{Cost\ per\\ Muffin}} = 14$$

$$2\times 3+2\times 4=2(3+4)$$

Example 1.151
Rishabh has b muffins. Radha has four m muffins. If each muffin costs two dollars, find the total cost of the muffins.

$$2(b+m)=2b+2m$$

Example 1.152: Working with Numbers
A. $2\times 49=98$
B. $2\times 449=998$

$$2\times 49=2(50-1)=2\times 50-2\times 1=100-2=98$$

$$2\times 449=2(500-1)=2\times 500-2\times 1=1000-2=998$$

For Variables

$$2(3x+5y)=(3x+5y)+(3x+5y)=6x+10y$$

$$2(3x+5y)=2\times 3x+2\times 5y=6x+10y$$

Example 1.153
Use the distributive property to expand the following expressions:

Integers

A. $3(2x+7z)$
B. $4(5p-7q)$
C. $5(-5r+7s)$
D. $3(-2y+5r)$
E. $4(-15-3p)-4(-p+5)$

Decimals

F. $2(0.2x + 0.3y)$
G. $0.2(0.2x + 0.3y)$

Fractions

H. $2\left(-\frac{1}{2}y + \frac{1}{5}z\right)$
I. $3\left(\frac{2}{5}a - \frac{2}{7}z\right)$
J. $-\frac{6}{7}\left(\frac{3}{4}j - \frac{5}{11}k\right)$

Integers

$$3(2x + 7z) = 3 \times 2x + 3 \times 7z = 6x + 21z$$
$$4(5p - 7q) = 20p - 28q$$
$$5(-5r + 7s) = -25r + 35s$$
$$3(-2y + 5r) = -6y + 15r$$

Decimals

Fractions

$$2\left(-\frac{1}{2}y + \frac{1}{5}z\right) = -y = \frac{2}{5}z$$
$$3\left(\frac{2}{5}a - \frac{2}{7}z\right) = \frac{6}{5}a - \frac{6}{7}z$$
$$-\frac{6}{7}\left(\frac{3}{4}j - \frac{5}{11}k\right) =$$

Example 1.154

$$4(-15 - 3p) - 4(-p + 5)$$

$$-60 - 12p + 4p - 20 =$$

Example 1.155

A. $-4 + y$
B. $-(4 + y)$

$$-4$$

Simplifying Expressions

Example 1.156

C. $7y + (3 - y)$
D. $7y - (3 - y)$

We need to add the expression inside the parentheses to the expression inside the parentheses. Note that there is no change to the expression:

$$7y + (3 - y) = 7y + 3 - y = 6y - 3$$

Here, we need to subtract the expression inside the parentheses. Hence, the sign of the terms inside the parentheses will change:

$$7y-(3-y)=7y-3-(-y)=7y-3+y=8y-3$$

Example 1.157

A. $5x-2x+4$
B. $5x-(2x+4)$

Example 1.158

C. $3x+2-(4x-5)$
D. $-2x+3-(4-5x)$

$$5x-(2x+4)=5x-2x-4=3x-4$$

$$3x+2-(4x-5)=3x+2-4x+5=-x+7$$
$$-2x+3-(4-5x)=-2x+3-4+5x=3x-1$$

Distributive Property with Fractions and Decimals

Example 1.159

$$0.3x+0.65-(0.4x-0.45)=0.3x+0.65-0.4x+0.45=-0.1x+1.1$$

$$3.4x-2.3-(4.3x-3.2)=3.4x-2.3-4.3+3.2=-0.9x+0.9$$

Example 1.160

$$\frac{1}{2}x+\frac{3}{4}-\left(\frac{2}{3}x-\frac{2}{5}\right)=\frac{1}{2}x+\frac{3}{4}-\frac{2}{3}x+\frac{2}{5}=-\frac{1}{6}x+\frac{23}{20}$$

Example 1.161

Use the distributive property to expand the following expressions, and then simplify:

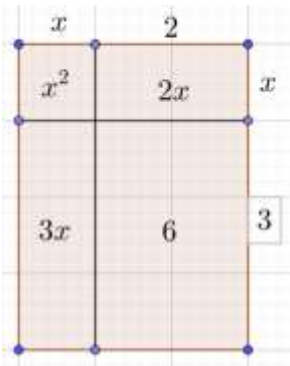
A. $2(3x+4y)+3(4x+y)$
B. $3(4x+y)-5(2x+6y)$
C. $5(5x+7y)-3(4x-y)$
D. $7(x-y)-3(8x-y)$

Example 1.162

Use the distributive property to expand the following expressions, and then simplify:

A. $0.25(8x+12y)$
B. $0.2(15p-10q)$
C. $-0.5(8a-7q)$
D. $-0.75(-16a-12b)$

The main idea here to simplify the calculations is that the decimals convert into fractions.



1.7 Binomial Expansions

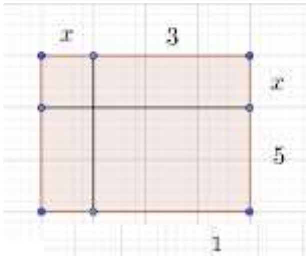
A. Multiplying Binomials

Example 1.163: Modelling with Rectangles

$$(x+2)(x+3)$$

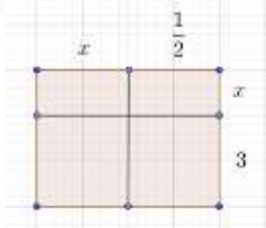
$$= x(x+3) + 2(x+3)$$
$$= x^2 + 3x + 2x + 6$$
$$= x^2 + 5x + 6$$

$$(x+3)(x+5)$$



$$= x(x+5) + 3(x+5)$$
$$= x^2 + 5x + 3x + 15$$
$$= x^2 + 8x + 15$$

$$\left(x+\frac{1}{2}\right)(x+3)$$



$$x(x+3) + \frac{1}{2}(x+3)$$
$$= x^2 + 3x + \frac{1}{2}x + \frac{3}{2}$$
$$= x^2 + \frac{7}{2}x + \frac{3}{2}$$

$$(2x+1)(x+3)$$

$$(\mathbf{2x} + \mathbf{1})(x+3)$$

Use the distributive property:

$$= \mathbf{2x}(x+3) + \mathbf{1}(x+3)$$
$$= 2x^2 + 6x + x + 3$$
$$= 2x^2 + 7x + 3$$

Example 1.164

A. $(x+2)(x-1)$
B. $(x-4)(x+2)$

Part A

$$(x+2)(x-1)$$
$$= x(x-1) + 2(x-1)$$
$$= x^2 - x + 2x - 2$$
$$= x^2 + x - 2$$

Part B

$$(x-4)(x+2)$$
$$= x(x+2) - 4(x+2)$$
$$= x^2 + 2x - 4x - 8$$
$$= x^2 - 2x - 8$$

Example 1.165

A. $(3x-1)(x+2)$
B. $\left(x-\frac{2}{3}\right)\left(x+\frac{1}{5}\right)$

Part A

$$\begin{aligned}&(3x-1)(x+2)\\&=3x(x+2)-1(x+2)\\&=3x^2+6x-x-2\\&=3x^2+5x-2\end{aligned}$$

Part B

$$\begin{aligned}&\left(x-\frac{2}{3}\right)\left(x+\frac{1}{5}\right)\\&=x\left(x+\frac{1}{5}\right)-\frac{2}{3}\left(x+\frac{1}{5}\right)\\&=x^2+\frac{1}{5}x-\frac{2}{3}x-\frac{2}{15}\\&=x^2+\frac{3}{15}x-\frac{10}{5}x-\frac{2}{15}\\&=x^2-\frac{7}{15}x-\frac{2}{15}\end{aligned}$$

B. Perfect Squares

1.166: Square of a Sum

$$(a+b)^2=a^2+2ab+b^2$$

$$\begin{aligned}&(a+b)^2\\&=(a+b)(a+b)\\&=a(a+b)+b(a+b)\\&=a^2+ab+ba+b^2\\&=a^2+2ab+b^2\end{aligned}$$

Example 1.167

A. $(x+2)^2$
B. $(x+3)^2$
C. $\left(x+\frac{1}{2}\right)^2$
D. $(2x+4)^2$
E. $\left(3x+\frac{2}{3}\right)^2$

$$(x+2)^2=x^2+4x+4$$
$$(x+3)^2=x^2+6x+9$$
$$\left(x+\frac{1}{2}\right)^2=x^2+(2)(x)\left(\frac{1}{2}\right)+\left(\frac{1}{2}\right)^2=x^2+x+\frac{1}{4}$$
$$(2x+4)^2=(2x)^2+(2)(2x)(4)+4^2=4x^2+16x+16$$
$$\left(3x+\frac{2}{3}\right)^2=(3x)^2+(2)(3x)\left(\frac{2}{3}\right)+\left(\frac{2}{3}\right)^2=9x^2+4x+\frac{4}{9}$$

1.168: Square of a Difference

$$(a-b)^2=a^2-2ab+b^2$$

$$(a-b)^2$$

$$= (a-b)(a-b)$$
$$= a(a-b) - b(a-b)$$
$$= a^2 - ab - ba + b^2$$
$$= a^2 - 2ab + b^2$$

Example 1.169

A. $(x-2)^2$
B. $(x-5)^2$
C. $\left(x-\frac{1}{3}\right)^2$
D. $(2x-3)^2$
E. $\left(5x-\frac{4}{7}\right)^2$

$$(x-2)^2 = x^2 - (2)(x)(2) + 2^2 = x^2 - 4x + 4$$
$$(x-5)^2 = x^2 - (2)(x)(5) + 25 = x^2 - 10x + 25$$
$$\left(x-\frac{1}{3}\right)^2 = x^2 - (2)(x)\left(\frac{1}{3}\right) + \left(\frac{1}{3}\right)^2 = x^2 - \frac{2}{3}x + \frac{1}{9}$$
$$(2x-3)^2 = (2x)^2 - (2)(2x)(3) + (-3)^2 = 4x^2 - 12x + 9$$
$$\left(5x-\frac{4}{7}\right)^2 = (5x)^2 - (2)(5x)\left(\frac{4}{7}\right) + \left(-\frac{4}{7}\right)^2 = 25x^2 - \frac{40}{7}x + \frac{16}{49}$$

C. Difference of Squares

1.170: Difference of Squares

$$(a+b)(a-b) = a^2 - b^2$$

Use the distributive property:

$$a(a-b) + b(a-b) = a^2 \boldsymbol{- ab + ab} - b^2 = a^2 - b^2$$

Example 1.171

A. $(x+3)(x-3)$
B. $(y+5)(y-5)$
C. $\left(p+\frac{1}{2}\right)\left(p-\frac{1}{2}\right)$
D. $\left(q+\frac{2}{3}\right)\left(q-\frac{2}{3}\right)$
E. $\left(r+\frac{4}{7}\right)\left(r-\frac{4}{7}\right)$
F. $(2a+3)(2a-3)$
G. $(5b+4)(5b-4)$
H. $(3c+1)(3c-1)$
I. $\left(6m+\frac{2}{3}\right)\left(6m-\frac{2}{3}\right)$
J. $\left(4n+\frac{5}{7}\right)\left(4n-\frac{5}{7}\right)$

$$(x+3)(x-3) = x^2 - 9$$
$$(y+5)(y-5) = y^2 - 25$$
$$\left(p+\frac{1}{2}\right)\left(p-\frac{1}{2}\right) = p^2 - \frac{1}{4}$$

$$\left(q+\frac{2}{3}\right)\left(q-\frac{2}{3}\right)=q^2-\frac{4}{9}$$
$$\left(r+\frac{4}{7}\right)\left(r-\frac{4}{7}\right)=r^2-\frac{16}{49}$$
$$(2a+3)(2a-3)=4a^2-9$$
$$(5b+4)(5b-4)=25b^2-16$$
$$(3c+1)(3c-1)=9c^2-1$$
$$\left(6m+\frac{2}{3}\right)\left(6m-\frac{2}{3}\right)=36m^2-\frac{4}{9}$$
$$\left(4n+\frac{5}{7}\right)\left(4n-\frac{5}{7}\right)=16n^2-\frac{25}{49}$$

Example 1.172

A. $(x^2+4)(x^2-4)$
B. $(y^3+3)(y^3-3)$
C. $(z^5+10)(z^5-10)$
D. $\left(p^4+\frac{1}{2}\right)\left(p^4-\frac{1}{2}\right)$
E. $\left(q^{12}+\frac{3}{4}\right)\left(q^{12}-\frac{3}{4}\right)$
F. $\left(r^9+\frac{11}{13}\right)\left(r^9-\frac{11}{13}\right)$

$$(x^2+4)(x^2-4)=x^4-16$$
$$(y^3+3)(y^3-3)=(y^3)^2-9=y^6-9$$
$$(z^5+10)(z^5-10)=z^{10}-100$$
$$\left(p^4+\frac{1}{2}\right)\left(p^4-\frac{1}{2}\right)=p^8-\frac{1}{4}$$
$$\left(q^{12}+\frac{3}{4}\right)\left(q^{12}-\frac{3}{4}\right)=q^{24}-\frac{9}{16}$$
$$\left(r^9+\frac{11}{13}\right)\left(r^9-\frac{11}{13}\right)=r^{18}-\frac{121}{169}$$

Example 1.173

A. $(2n+1)(2n-1)(n+5)$

$$(4n^2-1)(n+5)=4n^3+20n^2-n-5$$

1.8 Factoring

A. Basics

1.174: Multiplying by Zero

Multiplying anything by zero gives you zero. For any value of a:

$$a\times 0=0$$

Example 1.175

A. $3\times\frac{1}{4}\times 0\times\frac{5}{23}-8+2$

$$3 \times \frac{1}{4} \times 0 \times \frac{5}{23} - 8 + 2 = \underbrace{3 \times \frac{1}{4} \times 0 \times \frac{5}{23}}_{=0} \underbrace{-8 + 2}_{\neq 0} = 0 - 6 = -6$$

Example 1.176

A. Shyam took a number. Then he added five to the number. Then he multiplied it by twelve. Then he subtracted six. Then, he multiplied it by zero. Then, he divided it by eight. What was his final answer?

$$\frac{(x+5) \times 12 \times 0}{8} = \frac{0}{8} = 0$$

1.177: Creating a zero

$$5 \times 2 = 10 \Rightarrow One\ Zero$$
$$4 \times 25 = 100 \Rightarrow Two\ Zeros$$
$$8 \times 125 = 1000 \Rightarrow Three\ Zeros$$

Example 1.178

A. $2 \times 3 \times 5 \times 7 \times 4 \times 2 \times 25$
B. $6 \times 5 \times 4 \times 3 \times 2 \times 1$

Rearrange to put 2 with 5, and 4 with 25:

$$\underbrace{2 \times 5}_{=10} \times 3 \times 7 \times 2 \times \underbrace{4 \times 25}_{=100} = 42 \times 1000 = 42000$$

Example 1.179

A. Find the product of the first six natural numbers.
B. Find the product of the first six whole numbers.

Part A

$$6 \times 5 \times 4 \times 3 \times 2 \times 1 = 10 \times \underbrace{6 \times \underbrace{4 \times 3}_{12}}_{72} = 720$$

Part B

$$5 \times 4 \times 3 \times 2 \times 1 \times 0 = 0$$

B. Factoring

Factoring is the opposite of the distributive property.
We use the distributive property to open a bracket. In factoring, we create a bracket.

$$2x + 4y = \underbrace{2 \times x + 2 \times 2y}_{2\ is\ a\ common\ factor} = \underbrace{2(x + 2y)}_{\substack{Combine\ the\ common\ factor \\ using\ factoring}}$$

We can do a few more examples that fit the same pattern:

$$3x + 6y = 3 \times x + 3 \times 2y = 3(x + 2y)$$
$$6 \times 14 + 6 \times 36 = 6(14 + 36) = 6(50) = 300$$
$$4 \times 42 + 4 \times 58 = 4(42 + 58) = 4(100) = 400$$

Example 1.180

Factor completely:

A. $9x + 6y$

B. $4x + 12y$
C. $10x + 20y$
D. $5x + 25y$
E. $16x + 48y$
F. $32x + 64y$

$$HCF(9,6) = 3 \Rightarrow 4x + 12y = 4(x + 3y)$$
$$9x + 6y = 3(3x + 2y)$$
$$10x + 20y = 10(x + 2y)$$
$$5x + 25y = 5(x + 5y)$$
$$16x + 48y = 16(x + 3y)$$
$$32x + 64y = 32(x + 2y)$$

Example 1.181

A. $3x + 6y + 9z$
B. $12x + 24y + 36z$
C. $12x + 20y + 24z$
D. $35p + 63q + 210r$

$$3x + 6y + 9z = 3(x + 2y + 3z)$$
$$12x + 24y + 36z = 12(x + 2y + 3z)$$
$$12x + 20y + 24z =$$
$$35p + 63q + 210r =$$

C. Cancellation

Factoring can help us “cancel”. Once we factor, the numbers add up to zero.

Example 1.182

$$317 \times 5 + 317 \times (-5)$$

$$\mathbf{317} \times \mathbf{5} + \mathbf{317} \times (-\mathbf{5}) = \mathbf{317}(\mathbf{5} - \mathbf{5}) = \mathbf{317} \times \mathbf{0} = \mathbf{0}$$
$$\mathbf{317} \times \mathbf{5} + \mathbf{317} \times (-\mathbf{5}) = \mathbf{5}(317 - 317) = \mathbf{5}(0) = \mathbf{0}$$

D. Reducing multiplication to addition

The power of this technique really comes through when you have large numbers. We can reduce the chances of making a mistake by using factoring:

Example 1.183

$$572 \times 3 + 572 \times 7 = 572(3 + 7) = 572 \times 10 = 5720$$

$$\mathbf{572} \times 3 + \mathbf{572} \times 7 = \mathbf{572}(3 + 7) = \mathbf{572} \times 10 = 5720$$

Practice 1.184

Simplify:

A. $43 \times 4 + 43 \times 6$
B. $6 \times 43 + 6 \times 57$
C. $23 \times 2 + 23 \times 8$
D. $5 \times 99 + 5 \times 1$

$$43 \times 4 + 43 \times 6 = 43(4 + 6) = 43 \times 10 = 430$$

$$6 \times 43 + 6 \times 57 = 6(43 + 57) = 6 \times 100 = 600$$
$$23 \times 2 + 23 \times 8 = 23(2 + 8) = 23 \times 10 = 230$$
$$5 \times 99 + 5 \times 1 = 5(99 + 1) = 5 \times 100 = 500$$

E. Commutative Property

More difficult questions may not give us the expression in a readily useable form. We may need to manipulate it to make it suitable for factoring.

Example 1.185

$$3 \times 44 + 56 \times 3$$

It is not necessary that the common factor should be first in every set of numbers being multiplied. Because of the commutative property, we can rearrange it:

$$3 \times 44 + 56 \times 3 = 3 \times 44 + 3 \times 56 = 3(44 + 56) = 3(100) = 300$$

F. Multiplicative Identity

Pay attention to the next two examples carefully. We use the multiplicative identity to create a common factor. It's easy to make a mistake here, and put zero, so remember this one.

Example 1.186

$$34 \times 49 + 34$$

$$\mathbf{34} \times \mathbf{49} + \mathbf{34} \times \mathbf{1} = \mathbf{34}(\mathbf{49} + \mathbf{1}) = 34 \times 50$$

We could carry out the multiplication, but let us look at one more trick:

$$34 \times 50 = 17 \times 2 \times 50 = 17 \times 100 = 1700$$

Example 1.187

$$43 \times 99 + 44$$

Sometimes, the number to be factored is not so readily available. The first term is a multiple of 43, but the second term is not. However, we can break up the 44, as shown:

$$43 \times 99 + \underbrace{44}_{43+1}$$

Rewrite the 43 as 43×1:

$$\underbrace{43 \times 99 + 43 \times 1}_{\substack{\textit{Take 43 common from} \\ \textit{both terms}}} + \underbrace{1}_{\substack{\textit{Nothing to take} \\ \textit{common here}}}$$

Now the first two terms above have a 43. Take 43 common from them. The last term is not a multiple of 43. Leave it alone.

$$\underbrace{43(99 + 1)}_{\substack{\textit{After taking 43} \\ \textit{common}}} + \underbrace{1}_{\substack{\textit{Left as is} \\ \textit{Very important}}}$$

Now simplify as before:

$$43 \times 100 + 1 = 4300 + 1 = 4301$$

G. Simplifying and Rearranging

We can factor out a common factor with three or more terms as well.

Example 1.188

Simplify:

A. $3 \cdot 12 - 3 \cdot 6 + 3 \cdot 14$
B. $6 \cdot 2 + 25 \cdot 6 + 11 \cdot 6 + 6 \cdot 6$

$$3 \times 12 - 3 \times 6 + 3 \times 14 = 3(12 - 6 + 14) = 3(20) = 60$$
$$6 \cdot 2 + 25 \cdot 6 + 11 \cdot 6 + 6 \cdot 6 = 6(2 + 25 + 11 + 6) = 6(44) = \underbrace{240}_{6\times40} + \underbrace{24}_{6\times4} = 264$$

Example 1.189

Simplify:

$$99 + 88 - 77 + 66 - 55 + 44 - 33 + 22 - 11 + 0 - 121$$

$$99 + 88 - 77 + 66 - 55 + 44 - 33 + 22 - 11 + 0 - 121$$

Rewrite the terms as multiples of 11:

$$9 \times 11 + 8 \times 11 - 7 \times 11 + 6 \times 11 - 5 \times 11 + 4 \times 11 - 3 \times 11 + 2 \times 11 - 1 \times 11 - 11 \times 11$$

Take 11 Common:

$$11(9 + 8 - 7 + 6 - 5 + 4 - 3 + 2 - 1 - 11)$$

Form Pairs

$$11\left[9 + \underbrace{8 - 7}_{=1} + \underbrace{6 - 5}_{=1} + \underbrace{4 - 3}_{=1} + \underbrace{2 - 1}_{=1} - 11\right] = 11(9 + 1 + 1 + 1 + 1 + 1 - 11) = 11(3) = 33$$

Example 1.190

$$63 \cdot 15 + 5 \cdot 63 \cdot 10 + 7 \cdot 5 \cdot 63$$

It can be necessary to simplify the other parts of the expression before or after the factoring.

$$63 \cdot 15 + 50 \cdot 63 + 35 \cdot 63 = 63(15 + 50 + 35) = 63(100) = 6300$$

Example 1.191

$$12 \cdot 17 + 24 \cdot 7$$

It can be necessary or useful to rearrange numbers in order to make the calculation easier.

$$12 \cdot 17 + 24 \cdot 7 = 12 \cdot 17 + \underbrace{12 \cdot 2 \cdot 7}_{\substack{Create\ a\ 12 \\ Note\ this\ Step}} = 12(17 + 14) = 12 \times 31 = 12 \cdot 30 + 12 = 372$$

Example 1.192

Simplify:

A. $23 \times 67 + 23 \times 33$
B. $12 \times 17 + 13 \times 12$

$$23 \times 67 + 23 \times 33 = 23 \times 100 = 2300$$
$$12 \times 17 + 13 \times 12 = 12(17 + 13) = 12 \times 30 = 360$$

H. Word Problems

Example 1.193

Carrie has seven apple trees, each with 32 apples. Fisher has three apple trees, of the same variety that Carrie has, and hence, the same number of apples on each tree. Find the total number of apples that they have together.

$$7 \times 32 + 3 \times 32 = 32(7+3) = 32 \times 10 = 320$$

Example 1.194

An orchard has 29 mango trees of the *Rajapuri* variety, and 29 mango trees of the *Alphonso* variety. Each *Rajapuri* tree has six branches, and each branch has eight mangoes. Each *Alphonso* tree has four branches, and each branch has thirteen mangoes. Find the total number of mangoes that both have.

$$\underbrace{29 \times 6 \times 8}_{Rajpuri} + \underbrace{29 \times 4 \times 13}_{Alphonso} = 29 \times 48 + 29 \times 52 = 29(48+52) = 2900$$

I. Exponents

Example 1.195

$$\frac{2^{10} - 2^8}{2^7 - 2^6}$$

$$\frac{2^{10} - 2^8}{2^7 - 2^6} = \frac{\mathbf{2^8} \times 2^2 - \mathbf{2^8} \times 1}{\mathbf{2^6} \times 2^1 - \mathbf{2^6} \times 1} = \frac{2^8(2^2 - 1)}{2^6(2^1 - 1)} = \frac{2^2(3)}{1} = 12$$

Example 1.196: HCF and LCM with Variables

To find the HCF of two algebraic expressions, we find the HCF of the coefficients, and the HCF of each variable separately.

Find the HCF and LCMof

A. $36a^2bc^3$ and $27ab^2c$
B. $36a^2bc^3$ and $27ab^2$
C. $x^2 + x$ and $4x + 4$

Part A

$$HCF(36,27) = 9$$
$$HCF = (a^2, a) = a$$
$$HCF(b, b^2) = b$$
$$HCF(c^3, c) = c$$

Then, we multiply each of the above HCF's that we got:

$$9 \times a \times b \times c = 9abc \Rightarrow Final\ Answer$$

$$LCM(36,27) = LCM(2^2 \times 3^2, 3^3) = 2^2 \times 3^3 = 4 \times 27 = 104$$
$$LCM(a^2, a) = a^2$$
$$LCM(b, b^2) = b^2$$
$$LCM(c^3, c) = c^3$$

Then, we multiply each of the above HCF's that we got:

$$104 \times a^2 \times b^2 \times c^3 = 104a^2b^2c^3 \Rightarrow Final\ Answer$$

Part B

$$HCF(36,27) = 9$$
$$HCF = (a^2, a) = a$$
$$HCF(b, b^2) = b$$
$$HCF(c^3, 1) = 1$$

Then, we multiply each of the above HCF's that we got:

$$9 \times a \times b = 9ab \Rightarrow Final\ Answer$$

Part C

$$x^2 + x = 1 \times x(x+1)$$
$$4x + 4 = 4(x+1)$$

$$LCM(1,4) = 4$$
$$LCM(x, 1 = x^0) = x$$
$$LCM(x+1, x+1) = x+1$$

$$4 \times x \times (x+1) = 4x(x+1) \Rightarrow Final\ Answer$$

Example 1.197

$$\frac{2}{x^2+x} - \frac{x}{4x+4} = \frac{2}{x(x+1)} - \frac{x}{4(x+1)}$$

Find the LCM of the denominators:

$$LCM\big(x(x+1), 4(x+1)\big) = 4x(x+1)$$

Make the denominators common:

$$\left(\frac{2}{x(x+1)} \times \frac{4}{4}\right) - \left(\frac{x}{4(x+1)} \times \frac{x}{x}\right) = \frac{8}{4x(x+1)} - \frac{x^2}{4x(x+1)}$$

Carry out the addition now that the denominators are common

$$\frac{8 - x^2}{4x(x+1)}$$

J. Variables

Example 1.198

Factor:

A. $3x + 9y$
B. $4p + 6q$
C. $6p + 8q + 2r$
D. $x + x^2$
E. $x + x^3$
F. $x^2 + x^3$
G. $4x + 6x^2$
H. $z + z^2 + z^3$
I. $p^2 + p^4 + p^5$
J. $3a + 9a^2 + 27a^3$

$$3x + 9y = 3(x + 3y)$$
$$4p + 6q = 2(2p + 3q)$$
$$6p + 8q + 2r = 2(3p + 4q + r)$$
$$x + x^2 = 1(x + x^2) = \frac{x}{x}(x + x^2) = x \times \frac{1}{x}(x + x^2) = x\left(\frac{x}{x} + \frac{x^2}{x}\right) = x(1+x)$$
$$x + x^3 = x(1 + x^2)$$
$$x^2 + x^3 = x^2(1 + x)$$
$$4x + 6x^2 = 2x(2 + 3x)$$
$$z + z^2 + z^3 = z(1 + z + z^2)$$
$$p^2 + p^4 + p^5 = p^2(1 + p^2 + p^3)$$

$$3a+9a^2+27a^3=3a(1+3a+9a^2)$$

Example 1.199
Reduced to lowest terms, $\frac{a^2-b^2}{ab}-\frac{ab-b^2}{ab-a^2}$ is equal to: **(AHSME 1950/4)**

Simplify the second term as $\frac{ab-b^2}{ab-a^2}=\frac{b(a-b)}{a(a-b)}=\frac{b}{a}$. Then, the required expression is:

$$\frac{a^2-b^2}{ab}-\frac{b}{a}=\frac{a^2-b^2-b^2}{ab}=\frac{a^2}{ab}=\frac{a}{b}$$

Example 1.200: Grouping
Factor

A. $y(y+3)+3(y+3)$
B. $z(z+2)+(z+2)$

Factor $y+3$ from each term:

$$y(\boldsymbol{y+3})+3(\boldsymbol{y+3})=(y+3)(y+3)$$

$$z(z+2)+(z+2)$$

$$z(z+2)+(z+2)=(z+2)(z+1)$$

Example 1.201: Variables in the Exponents
Factor:

A. 5^x+5^{x+3}
B. 3^x+3^{x+2}
C. $2^{x+1}+2^{x+2}$
D. $4^x+4^{x+\frac{1}{2}}$
E. $27^{x+1}+27^{x+\frac{1}{3}}$

$$5^x+5^{x+3}=\frac{5^x}{5^x}(5^x+5^{x+3})=5^x\left(\frac{5^x}{5^x}+\frac{5^{x+3}}{5^x}\right)=5^x(1+5^3)=5^x(1+125)=126\times 5^x$$

$$\textit{Shortcut}: 5^x+5^{x+3}=5^x(1+5^3)=5^x(1+125)=126\times 5^x$$

$$3^x(1+3^2)=3^x(1+9)=10\times 3^x$$
$$2^{x+1}+2^{x+2}=2^x(2^1+2^2)=2^x(2+4)=6\times 2^x=3\times 2^{x+1}$$
$$4^x\left(1+4^{\frac{1}{2}}\right)=4^x(1+2)=3\times 4^x=3\times 2^{2x}$$

K. Difference of Squares

Example 1.202
Factor completely:

A. $x^2 - 16$
B. $y^2 - 25$
C. $z^2 - 121$

Example 1.203
Factor completely:
A. $4x^2 - 36$
B. $3y^2 - 75$

$$4x^2 - 36 = 4(x^2 - 9) = 4(x+3)(x-3)$$

Example 1.204
Factor completely:
A. $x^4 - 1$
B. $y^4 - 81$
C. $z^4 - 16$

Example 1.205
Factor completely:
A. $3x^4 - 3$
B. $5y^4 - 80$

Example 1.206
Factor completely:
A. $x^8 - 1$

$$(x^4 + 1)(x^2 + 1)(x + 1)(x - 1)$$

1.9 Rational Expressions

A. Rational Expressions

Example 1.207

$$2x - 5 - \frac{x-8}{x+4}$$

$$\frac{2x-5}{1} - \frac{x-8}{x+4}$$

LCM is $x + 4$

$$\frac{(2x-5)(x+4) - (x-8)}{x+4}$$
$$\frac{2x^2 + 8x - 5x - 20 - x + 8}{x+4}$$
$$\frac{2x^2 + 2x - 12}{x+4}$$

Example 1.208

$$\frac{x-5}{5-x}$$

$$\frac{x-5}{5-x} = \frac{-(5-x)}{5-x} = -1$$

Rational Expression Values

Example 1.209

If x is a number between 0 and 10. (for example, x can be 1.01 or 0.75 or 2.08) and y is the number $x + \frac{1}{x}$ then can y be bigger than 2005? **(NMTC Primary/Final 2005/8)**

Since there is a fraction in the expression, consider values of x in two parts

Part A: Where the denominator is greater than or equal to 1

This means that x is greater than or equal to one. The value of the expression is made larger by making x larger. The largest value possible is

$$10 + \frac{1}{10} = 10 + 0.1 = 10.1$$

Part B: Where the denominator is less than 1

As the value of x decreases, the value of $\frac{1}{x}$ increases.

For example, consider values of x and $\frac{1}{x}$ below:

x	$\frac{1}{2}$	$\frac{1}{10}$	$\frac{1}{100}$	$\frac{1}{1000}$	$\frac{1}{10{,}000}$
$\frac{1}{x}$	2	10	100	1000	10,000

Example 1.210

Given that x is a positive number, what happens to the following expressions as x increases:

A. $\frac{3}{x}$
B. $\frac{x}{3}$
C. $\frac{3}{x+x}$
D. $\frac{x+3}{3}$

$$Decreases$$
$$Increases$$

Example 1.211

If in the formula $C = \frac{en}{R+nr}$, where e, n, R and r are all positive, n is increased while e, R and r are kept constant, then C:

A. Increases
B. Decreases

C. Remains constant
D. Increases and then decreases
E. Decreases and then increases (AHSME 1950/11)

$$C = \frac{en}{R+nr} = \frac{n}{\frac{R}{n}+r} \Rightarrow \frac{R}{n} \text{ decreases} \Rightarrow \text{Denominator Decreases} \Rightarrow \text{Expression Increases}$$

1.10 Further Topics

212 Examples